

PRINCIPLES *of* GENETICS

Fifth Edition

ROBERT H. TAMARIN

PREFACE

The twentieth century began with the rediscovery of Mendel's rules of inheritance and ended with the complete sequence of the human genome, one of the most monumental scientific accomplishments of all time. What lies in the future? What will the twenty-first century, the century of genomics, bring? Will geneticists a hundred years from now speak of a complete cure for cancer, heart disease, and mental illness? Will we have a cure for autoimmune diseases such as diabetes and arthritis? Will aging be slowed or even prevented? Will we have a complete understanding of the process of development and a concurrent elimination of birth defects and developmental problems? Will genetics put an end to world hunger? How will we live, and what will be the quality of our lives? The students who now are taking genetics will learn the answers to these questions as time progresses. Some students will contribute to the answers.

The science of genetics includes the rules of inheritance in cells, individuals, and populations and the molecular mechanisms by which genes control the growth, development, and appearance of an organism. No area of biology can truly be appreciated or understood without an understanding of genetics because genes not only control cellular processes, they also determine the course of evolution. Genetic concepts provide the framework for the study of modern biology.

This text provides a **balanced treatment** of the major areas of genetics in order to prepare the student for upper-level courses and to help share in the excitement of research. Most readers of this text will have taken a general biology course and will have had some background in cell biology and organic chemistry. For an understanding of the concepts in this text, however, the motivated student will need to have completed only an introductory biology course and have had some chemistry and algebra in high school.

Genetics is commonly divided into three areas: classical, molecular, and population, although molecular advancements have blurred these distinctions. Many genetics teachers feel that a historical approach provides a sound introduction to the field and that a thorough grounding in Mendelian genetics is necessary for an understanding of molecular and population genetics—an approach this text follows. Other teachers, however, may prefer to begin with molecular genetics. For this reason, the chapters have been grouped as **units that allow for flexibility**

in their use. A comprehensive glossary and index will help maintain continuity if the instructor chooses to change the order of the chapters from the original.

An understanding of genetics is crucial to advancements in medicine, agriculture, and many industries. Genetic controversies—such as the pros and cons of the Human Genome Project, the potential ethical and medical risks of recombinant DNA and cloning of mammals, and human behavioral genetic issues such as the degree of inheritance of homosexuality, alcoholism, and intelligence—have captured the interest of the general public. Throughout this text, we examine the **implications for human health and welfare** of the research conducted in universities and research laboratories around the world; boxed material in the text gives insight into genetic techniques, controversies, and breakthroughs.

Because genetics is the first analytical biology course for many students, some may have difficulty with its quantitative aspects. There is no substitute for work with pad and pencil. This text provides a larger number of problems to help the student learn and retain the material. All problems within the body of the text and a selection at the end of the chapters should be worked through as they are encountered. After the student has worked out the problems, he or she can refer to the answer section in Appendix A. We provide solved problems at the end of each chapter to help.

In this text, we stress **critical thinking**, an approach that emphasizes understanding over memorization, experimental proof over the pronouncements of authorities, problem solving over passive reading, and active participation in lectures. The latter is best accomplished if the student reads the appropriate text chapter before coming to lecture rather than after. That way the student can use the lecture to gain insight into difficult material rather than spending the lecture hectically transcribing the lecturer's comments onto the notebook page.

For those students who wish to pursue particular topics, a **reference section** in the back of the text provides chapter-by-chapter listings of review articles and articles in the original literature. Although some of these articles might be difficult for the beginner to follow, each is a landmark paper, a comprehensive summary, or a paper with some valuable aspect. Some papers may contain an insightful photograph or diagram. Some magazines and journals are especially recommended for the student to look at periodically, including *Scientific American*,

Science, and *Nature*, because they contain nontechnical summaries as well as material at the cutting edge of genetics. Some articles are included to help the instructor find supplementary materials related to the concepts in this book. Photographs of selected geneticists also are included. Perhaps the glimpse of a face from time to time will help add a human touch to this science.

The World Wide Web also can provide a valuable resource. The textbook has its own website: www.mhhe.com/tamarin7. In addition, the student can find much material of a supplemental nature by “surfing” the web. Begin with a search engine such as: www.yahoo.com, or www.google.com and type in a key word. Follow the links from there. Remember that the material on the web is “as is”; it includes a lot of misinformation. Usually, content from academic, industrial, and organizational sources is relatively reliable; however, caveat emptor—buyer beware. Often in surfing for scientific key words, the student will end up at a scientific journal or book that does not have free access. Check with the university librarian to see if access might be offered to that journal or book. The amount of information that is accurate and free is enormous. Be sure to budget the amount of time spent on the Internet.

NEW TO THIS EDITION

Since the last edition of this text, many exciting discoveries have been made in genetics. All chapters have been updated to reflect those discoveries. In particular:

- The chapter on Recombinant DNA Technology has been revised to be a chapter on Genomics, Biotechnology, and Recombinant DNA (sixth edition chapter 12 has become chapter 13 in this edition). The chapter includes new material on the completion of the Human Genome Project, bioinformatics, proteomics, and the latest techniques in creating cDNA and knockout mice.
- The chapter on Control of Transcription in Eukaryotes (sixth edition chapter 15 has become chapter 16 in this edition) has been completely reorganized and rewritten to emphasize signal transduction, specific transcription factors, methylation, and chromatin remodeling in control of gene expression; as in the last edition, there are specific sections on *Drosophila* and plant development, cancer, and immunogenetics.
- For better continuity, the chapter on Mutation, Recombination, and DNA Repair has been moved to follow the chapters on Transcription and Translation (sixth edition chapter 16 has become chapter 12 in this edition).

- The material in chapter 3 on Genetic Control of the Cell Cycle has been upgraded to a chapter section on the Cell Cycle.
- Molecular material throughout the book has been completely updated to include such subjects as numerous DNA repair polymerases and their functioning; base-flipping; TRAP control of attenuation; and chromosomes.

LEARNING AIDS FOR THE STUDENT

To help the student learn genetics, as well as enjoy the material, we have made every effort to provide pedagogical aids. These aids are designed to help organize the material and make it understandable to students.

- **Study Objectives** Each chapter begins with a set of clearly defined, page-referenced objectives. These objectives preview the chapter and highlight the most important concepts.
- **Study Outline** The chapter topics are provided in an outline list. These headings consist of words or phrases that clearly define what the various sections of the chapter contain.
- **Boldface Terms** Throughout the chapter, all new terms are presented in boldface, indicating that each is defined in the glossary at the end of the book.
- **Boxed Material** In most chapters, short topics have been set aside in boxed readings, outside the main body of the chapter. These boxes fall into four categories: **Historical Perspectives**, **Experimental Methods**, **Biomedical Applications**, and **Ethics and Genetics**. The boxed material is designed to supplement each chapter with entertaining, interesting, and relevant topics.
- **Full Color Art and Graphics** Many genetic concepts are made much clearer with full-color illustrations and the latest in molecular computer models to help the student visualize and interpret difficult concepts. We've added thirty new photographs and over a hundred new and modified line drawings to this edition.
- **Summary** Each chapter summary recaps the study objectives at the beginning of the chapter. Thus, the student can determine if he or she has gained an understanding of the material presented in the study objectives and reinforce them with the summary.
- **Solved Problems** From two to four problems are worked out at the end of each chapter to give the student practice in solving and understanding basic problems related to the material.
- **Exercises and Problems** At the end of the chapter are numerous problems to test the student's

understanding of the material. These problems are grouped according to the sections of the chapter. Answers to the odd-numbered problems are presented in Appendix A, with the even-numbered problems answered only in the Student Study Guide so that the student and instructor can be certain that the student is gaining an understanding of the material.

- **Critical Thinking Questions** Two critical thinking questions at the end of each chapter are designed to help the student develop an ability to evaluate and solve problems. The answer to the first critical thinking question can be found in Appendix A, and the answer to the second question is in the Student Study Guide.

ANCILLARY MATERIALS

For the Instructor

- **Website.** Visit us at www.mhhe.com/tamarin7. Here instructors will find jpeg files of the line drawings and tables suitable for downloading into PowerPoint, quizzes for study support, and links to genetic sites. In addition, instructors will also find a link to our hugely successful **PageOut: The Course Website Development Center**, where instructors can create a professional-looking, customized course website. It's incredibly easy to use, and you need not know html coding.
- **Visual Resource Library (VRL).** This Windows- and Macintosh-compatible CD-ROM has all the line drawings and tables from the text suitable for PowerPoint presentations. (ISBN 0072334266)
- **Instructor's Manual with Test Item File.** Available on the website, the Instructor's Manual contains outlines, key words, summaries, instructional hints, and supplemental aids. The Test Item File contains 35 to 50 objective questions with answers for each chapter. (ISBN 0072334215)
- **Test Item File on MicroTest III Classroom Testing Software** is an easy-to-use CD-ROM test generator also offered free upon request to adopters of this text. The software requires no programming experience and is compatible with Windows or Macintosh systems. (ISBN 0072334231).

For the Student

- **Website.** Visit us at www.mhhe.com/tamarin7. Here the student will find quizzes for study support, web exercises and resources, and links to genetic sites.
- **Genetics: From Genes to Genomes CD-ROM**, by Ann E. Reynolds, University of Washington. Packaged free with every text, this CD-ROM covers the most chal-

lenging concepts in the course and makes them more understandable through the presentation of full-color, narrated animations and interactive exercises. The text indicates related topics on the CD with the following icon:



- **Student Study Guide.** This study guide features key concepts, problem-solving hints, practice problems, terms, study questions, and answers to even-numbered questions in the text. (ISBN 0072334207)
- **Laboratory Manual of Genetics 4/e**, by A. M. Winchester and P. J. Wejksnora, University of Wisconsin-Milwaukee. This manual for the genetics laboratory features classical and molecular biology exercises that give students the opportunity to apply the scientific method to "real"—not simulated—lab investigations. (ISBN 0697122875)
- **Case Workbook in Human Genetics, 2/e**, by Ricki Lewis, SUNY-Albany. The Workbook includes thought-provoking case studies in human genetics, with many examples gleaned from the author's experiences as a practicing genetic counselor. (ISBN 0072325305) Also included is the Answer Key. (ISBN 0072439009)

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1

INTRODUCTION

STUDY OBJECTIVES

1. To examine a brief overview of the modern history of genetics 3
2. To gain an overview of the topics included in this book—the syllabus of genetics 4
3. To analyze the scientific method 5
4. To look at why certain organisms and techniques have been used preferentially in genetics research 7

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Chameleon, *Cameleo pardalis*.

(© Art Wolfe/Tony Stone Images.)

Genetics is the study of inheritance in all of its manifestations, from the distribution of human traits in a family pedigree to the biochemistry of the genetic material in our chromosomes—deoxyribonucleic acid, or DNA. It is our purpose in this book to introduce and describe the processes and patterns of inheritance. In this chapter, we present a broad outline of the topics to be covered as well as a summary of some of the more important historical advancements leading to our current understanding of genetics.

A BRIEF OVERVIEW OF THE MODERN HISTORY OF GENETICS

For a generation of students born at a time when incredible technological advances are commonplace, it is valuable to see how far we have come in understanding the mechanisms of genetic processes by taking a very brief, encapsulated look at the modern history of genetics. Although we could discuss prehistoric concepts of animal and plant breeding and ideas going back to the ancient Greeks, we will restrict our brief look to events beginning with the discovery of cells and microscopes. For our purposes, we divide this recent history into four periods: before 1860, 1860–1900, 1900–1944, and 1944 to the present.

Before 1860

Before 1860, the most notable discoveries paving the way for our current understanding of genetics were the development of light microscopy, the elucidation of the cell theory, and the publication in 1859 of Charles Darwin's *The Origin of Species*. In 1665, Robert Hooke coined the term *cell* in his studies of cork. Hooke saw, in fact, empty cells observed at a magnification of about thirty power. Between 1674 and 1683, Anton van Leeuwenhoek discovered living organisms (protozoa and bacteria) in rainwater. Leeuwenhoek was a master lens maker and produced magnifications of several hundred power from single lenses (fig. 1.1). More than a hundred years passed before compound microscopes could equal Leeuwenhoek's magnifications. In 1833, Robert Brown (the discoverer of Brownian motion) discovered the nuclei of cells, and between 1835 and 1839, Hugo von Mohl described mitosis in nuclei. This era ended in 1858, when Rudolf Virchow summed up the concept of the cell theory with his Latin aphorism *omnis cellula e cellula*: all cells come from preexisting cells. Thus, by 1858, biologists had an understanding of the continuity of cells and knew of the cell's nucleus.

1860–1900

The period from 1860 to 1900 encompasses the publication of Gregor Mendel's work with pea plants in 1866 to the rediscovery of his work in 1900. It includes the discoveries of chromosomes and their behavior—insights that shed new light on Mendel's research.

From 1879 to 1885, with the aid of new staining techniques, W. Flemming described the chromosomes—first noticed by C. von Nägeli in 1842—including the way they split during division, and the separation of sister chromatids and their movement to opposite poles of the dividing cell during mitosis. In 1888, W. Waldeyer first used the term *chromosome*. In 1875, O. Hertwig described the fusion of sperm and egg to form the zygote. In the 1880s, Theodor Boveri, as well as K. Rabl and E. van Breden, hypothesized that chromosomes are individual structures with continuity from one generation to the next despite their “disappearance” between cell divisions. In 1885, August Weismann stated that inheritance is based exclusively in the nucleus. In 1887, he predicted the occurrence of a reductional division, which we now call meiosis. By 1890, O. Hertwig and T. Boveri had described the process of meiosis in detail.

1900–1944

From 1900 to 1944, modern genetics flourished with the development of the chromosomal theory, which showed

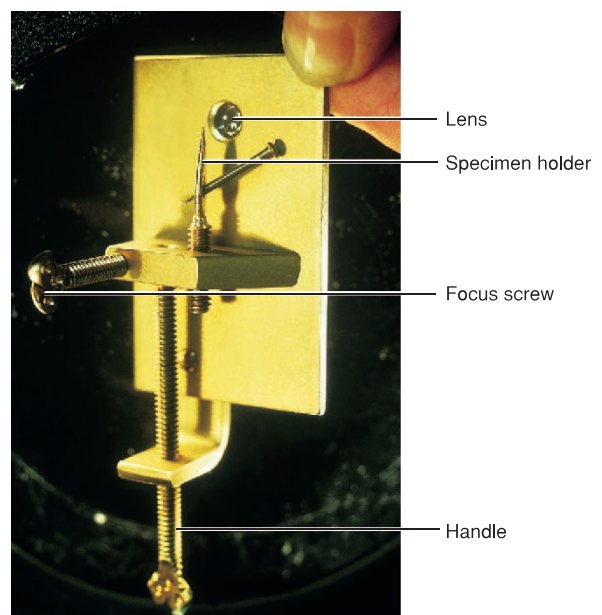


Figure 1.1 One of Anton van Leeuwenhoek's microscopes, ca. 1680. This single-lensed microscope magnifies up to 200x. (© Kathy Talaro/Visuals Unlimited, Inc.)

that chromosomes are linear arrays of genes. In addition, the foundations of modern evolutionary and molecular genetics were derived.

In 1900, three biologists working independently—Hugo de Vries, Carl Correns, and Erich von Tschermak—rediscovered Mendel's landmark work on the rules of inheritance, published in 1866, thus beginning our era of modern genetics. In 1903, Walter Sutton hypothesized that the behavior of chromosomes during meiosis explained Mendel's rules of inheritance, thus leading to the discovery that genes are located on chromosomes. In 1913, Alfred Sturtevant created the first genetic map, using the fruit fly. He showed that genes existed in a linear order on chromosomes. In 1927, L. Stadler and H. J. Muller showed that genes can be mutated artificially by X rays.

Between 1930 and 1932, R. A. Fisher, S. Wright, and J. B. S. Haldane developed the algebraic foundations for our understanding of the process of evolution. In 1943, S. Luria and M. Delbrück demonstrated that bacteria have normal genetic systems and thus could serve as models for studying genetic processes.

1944–Present

The period from 1944 to the present is the era of molecular genetics, beginning with the demonstration that DNA is the genetic material and culminating with our current explosion of knowledge due to recombinant DNA technology.

In 1944, O. Avery and colleagues showed conclusively that deoxyribonucleic acid—DNA—was the genetic material. James Watson and Francis Crick worked out the structure of DNA in 1953. Between 1968 and 1973, W. Arber, H. Smith, and D. Nathans, along with their colleagues, discovered and described restriction endonu-

cleases, the enzymes that opened up our ability to manipulate DNA through recombinant DNA technology. In 1972, Paul Berg was the first to create a recombinant DNA molecule.

Since 1972, geneticists have cloned numerous genes. Scientists now have the capability to create transgenic organisms, organisms with functioning foreign genes. For example, we now have farm animals that produce pharmaceuticals in their milk that are harvested easily and inexpensively for human use. In 1997, the first mammal was cloned, a sheep named Dolly. The sequence of the entire human genome was determined in 2000; we will spend the next century mining its information in the newly created field of genomics, the study of the complete genetic complement of an organism. Although no inherited disease has yet been cured by genetic intervention, we are on the verge of success in numerous diseases, including cancer.

The material here is much too brief to convey any of the detail or excitement surrounding the discoveries of modern genetics. Throughout this book, we will expand on the discoveries made since Darwin first published his book on evolutionary theory in 1859 and since Mendel was rediscovered in 1900.

THE THREE GENERAL AREAS OF GENETICS

Historically, geneticists have worked in three different areas, each with its own particular problems, terminology, tools, and organisms. These areas are classical genetics, molecular genetics, and evolutionary genetics. In *classical genetics*, we are concerned with the chromosomal theory of inheritance; that is, the concept that genes are

Table 1.1 The Three Major Areas of Genetics—Classical, Molecular, and Evolutionary—and the Topics They Cover

Classical Genetics	Molecular Genetics	Evolutionary Genetics
Mendel's principles	Structure of DNA	Quantitative genetics
Meiosis and mitosis	Chemistry of DNA	Hardy-Weinberg equilibrium
Sex determination	Transcription	Assumptions of equilibrium
Sex linkage	Translation	Evolution
Chromosomal mapping	DNA cloning and genomics	Speciation
Cytogenetics (chromosomal changes)	Control of gene expression	
	DNA mutation and repair	
	Extrachromosomal inheritance	

located in a linear fashion on chromosomes and that the relative positions of genes can be determined by their frequency in offspring. *Molecular genetics* is the study of the genetic material: its structure, replication, and expression, as well as the information revolution emanating from the discoveries of recombinant DNA techniques (genetic engineering, including the Human Genome Project). *Evolutionary genetics* is the study of the mechanisms of evolutionary change, or changes in gene frequencies in populations. Darwin's concept of evolution by natural selection finds a firm genetic footing in this area of the study of inheritance (table 1.1).

Today these areas are less clearly defined because of advances made in molecular genetics. Information coming from the study of molecular genetics allows us to understand better the structure and functioning of chromosomes on the one hand and the mechanism of natural selection on the other. In this book we hope to bring together this information from a historical perspective. From Mendel's work in discovering the rules of inheritance (chapter 2) to genetic engineering (chapter 13) to molecular evolution (chapter 21), we hope to present a balanced view of the various topics that make up genetics.

HOW DO WE KNOW?

Genetics is an empirical science, which means that our information comes from observations of the natural world. The *scientific method* is a tool for understanding these observations (fig. 1.2). At its heart is the experiment, which tests a guess, called a hypothesis, about how something works. In a good experiment, only two types of outcomes are possible: outcomes that support the hypothesis and outcomes that refute it. Scientists say these outcomes provide *strong inference*.

For example, you might have the idea that organisms can inherit acquired characteristics, an idea put forth by Jean-Baptiste Lamarck (1744–1829), a French biologist. Lamarck used the example of short-necked giraffes evolving into the long-necked giraffes we know of today. He suggested that giraffes that reached higher into trees to get at edible leaves developed longer necks. They passed on these longer necks to their offspring (in small increments in each generation), leading to today's long-necked giraffes. An alternative view, *evolution by natural selection*, was put forward in 1859 by Charles Darwin. According to the Darwinian view, giraffes normally varied in neck length, and these variations were inherited. Giraffes with slightly longer necks would be at an advantage in reaching edible leaves in trees. Therefore, over

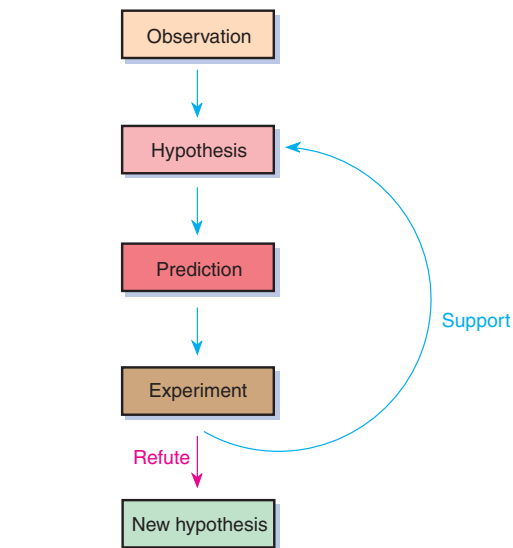


Figure 1.2 A schematic of the scientific method. An observation leads the researcher to propose a hypothesis, and then to make predictions from the hypothesis and to test these predictions by experiment. The results of the experiment either support or refute the hypothesis. If the experiment refutes the hypothesis, a new hypothesis must be developed. If the experiment supports the hypothesis, the researcher or others design further experiments to try to disprove it.

time, the longer-necked giraffes would survive and reproduce better than the shorter-necked ones. Thus, longer necks would come to predominate. Any genetic *mutations* (changes) that introduced greater neck length would be favored.

To test Lamarck's hypothesis, you might begin by designing an experiment. You could do the experiment on giraffes to test Lamarck's hypothesis directly; however, giraffes are difficult to acquire, maintain, and breed. Remember, though, that you are testing a general hypothesis rather than a specific hypothesis about giraffes. Thus, if you are clever enough, you can test the hypothesis with almost any organism. You would certainly choose one that is easy to maintain and manipulate experimentally. Later, you can verify the generality of any particular conclusions with tests on other organisms.

You might decide to use lab mice, which are relatively inexpensive to obtain and keep and have a relatively short generation time of about six weeks, compared with the giraffe's gestation period of over a year. Instead of looking at neck length, you might simply cut off the tip of the tail of each mouse (in a painless manner), using shortened tails as the acquired characteristic. You could then

BOX 1.1

As the pictures of geneticists throughout this book indicate, science is a very human activity; people living within societies explore scientific ideas and combine their knowledge. The society in which a scientist lives can affect not only how that scientist perceives the world, but also what that scientist can do in his or her scholarly activities. For example, the United States and other countries decided that mapping the entire human genome would be valuable (see chapter 13). Thus, granting agencies have directed money in this direction. Since much of scientific research is expensive, scientists often can only study areas for which funding is available. Thus, many scientists are working on the Human Genome Project. That is a positive example of society directing research. Examples also exist in which a societal decision has had negative consequences for both the scientific establishment and the society itself. An example is

Ethics and Genetics

The Lysenko Affair

the Lysenko affair in the former Soviet Union during Stalin's and Krushchev's reigns.

Trofim Denisovich Lysenko was a biologist in the former Soviet Union researching the effects of temperature on plant development. At the same time, the preeminent Soviet geneticist was Nikolai Vavilov. Vavilov was interested in improving Soviet crop yields by growing and mating many varieties and selecting the best to be the breeding stock of the next generation. This is the standard way of improving a plant crop or livestock breed (see chapter 18, "Quantitative Inheritance"). The method conforms to genetic principles and therefore is successful. However, it is a slow process that only gradually improves yields.

Lysenko suggested that crop yields could be improved quickly by the inheritance of acquired characteristics (see chapter 21, "Evolution and Speciation"). Although doomed to fail because they denied the true and correct mechanisms of inheritance, Lysenko's ideas were greeted with much enthusiasm by the political elite. The enthusiasm was due not only to the fact that Lysenko promised immediate improvements in crop yields, but also to the fact that Lysenkoism was politically favored. That is, Lysenkoism fit in very well with communism; it promised that nature could be manipulated easily and immediately. If people could manipulate nature so easily, then communism could easily convert people to its doctrines.

Not only did Stalin favor Lysenkoism, but Lysenko himself was favored politically over Vavilov because Lysenko came from peasant stock, whereas Vavilov was from a wealthy family. (Remember that communism

mate these short-tailed mice to see if their offspring have shorter tails. If they do not, you could conclude that a shortened tail, an acquired characteristic, is not inherited. If, however, the next generation of mice have tails shorter than those of their parents, you could conclude that acquired characteristics can be inherited.

One point to note is that every good experiment has a *control*, a part of the experiment that ensures that some unknown variable, often specific to a particular time and place, is not causing the observed changes. For example, in your experiment, the particular food the mice ate may have had an effect on their growth, resulting in offspring with shorter tails. To control for this, you could handle a second group of mice in the exact same way that the experimental mice are handled, except you would not cut off their tails. Any reduction in the lengths of the tails of the offspring of the control mice would indicate an artifact of the experiment rather than the inheritance of acquired characteristics.

The point of doing this experiment (with the control group), as trivial as it might seem, is to determine the an-

swer to a question using data based on what happens in nature. If you design your experiment correctly and carry it out without error, you can be confident about your results. If your results are negative, as ours would be here, then you would reject your hypothesis. Testing hypotheses and rejecting those that are refuted is the essence of the scientific method.

In fact, most of us live our lives according to the scientific method without really thinking about it. For example, we know better than to step out into traffic without looking because we are aware, from experience (observation, experimentation), of the validity of the laws of physics. Although from time to time anti-intellectual movements spread through society, few people actually give up relying on their empirical knowledge of the world to survive (box 1.1).

Nothing in this book is inconsistent with the scientific method. Every fact has been gained by experiment or observation in the real world. If you do not accept something said herein, you can go back to the *original literature*, the published descriptions of original experi-

was a revolution of the working class over the wealthy aristocracy.) Supported by Stalin, and then Krushchev, Lysenko gained inordinate power in his country. All visible genetic research in the former Soviet Union was forced to conform to Lysenko's Lamarckian views. People who disagreed with him were forced out of power; Vavilov was arrested in 1940 and died in prison in 1943. It was not until Nikita Krushchev lost power in 1964 that Lysenkoism fell out of favor. Within months, Lysenko's failed pseudoscience was repudiated and Soviet genetics got back on track.

For thirty years, Soviet geneticists were forced into fruitless endeavors, forced out of genetics altogether, or punished for their heterodox views. Superb scientists died in prison while crop improvement programs failed, all because the Soviet dictators favored Lysenkoism. The message of this affair is clear: Politicians can support research that agrees with their political agenda and punish scientists



Trofim Denisovich Lysenko (1898–1976) shows branched wheat to collective farmers in the former Soviet Union. (© SOVFOTO.)

doing research that disagrees with this agenda, but politicians cannot change the truth of the laws of nature. Science, to be effective, must be

done in a climate of open inquiry and free expression of ideas. The scientific method cannot be subverted by political bullies.

ments in scientific journals (as cited at the end of the book) and read about the work yourself. If you still don't believe a conclusion, you can repeat the work in question either to verify or challenge it. This is in keeping with the nature of the scientific method.

As mentioned, the results of experimental studies are usually published in scientific journals. Examples of journals that many geneticists read include *Genetics*, *Proceedings of the National Academy of Sciences*, *Science*, *Nature*, *Evolution*, *Cell*, *American Journal of Human Genetics*, *Journal of Molecular Biology*, and hundreds more. The reported research usually undergoes a process called *peer review* in which other scientists review an article before it is published to ensure its accuracy and its relevance. Scientific articles usually include a detailed justification for the work, an outline of the methods that allows other scientists to repeat the work, the results, a discussion of the significance of the results, and citations of prior work relevant to the present study.

At the end of this book, we cite journal articles describing research that has contributed to each chapter.

(In chapter 2, we reprint part of Gregor Mendel's work, and in chapter 9, we reprint a research article by J. Watson and F. Crick in its entirety.) We also cite secondary sources, that is, journals and books that publish syntheses of the literature rather than original contributions. These include *Scientific American*, *Annual Review of Biochemistry*, *Annual Review of Genetics*, *American Scientist*, and others. You are encouraged to look at all of these sources in your efforts both to improve your grasp of genetics and to understand how science progresses.

WHY FRUIT FLIES AND COLON BACTERIA?

As you read this book, you will see that certain organisms are used repeatedly in genetic experiments. If the goal of science is to uncover generalities about the living world, why do geneticists persist in using the same few organisms

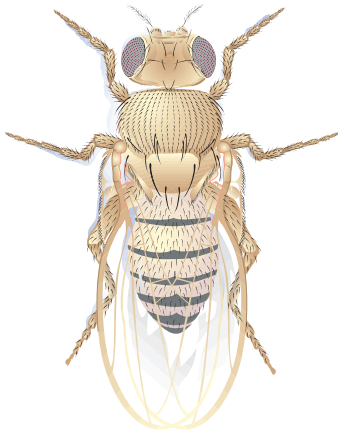


Figure 1.3 Adult female fruit fly, *Drosophila melanogaster*. Mutations of eye color, bristle type and number, and wing characteristics are easily visible when they occur.

in their work? The answer is probably obvious: the organisms used for any particular type of study have certain attributes that make them desirable *model organisms* for that research.

In the early stages of genetic research, at the turn of the century, no one had yet developed techniques to do genetic work with microorganisms or mammalian cells. At that time, the organism of preference was the fruit fly, *Drosophila melanogaster*, which developmental biologists had used (fig. 1.3). It has a relatively short generation time of about two weeks, survives and breeds well in the lab, has very large chromosomes in some of its cells, and has many aspects of its *phenotype* (appearance) genetically controlled. For example, it is easy to see the external results of mutations of genes that control eye color, bristle number and type, and wing characteristics such as shape or vein pattern in the fruit fly.

At the middle of this century, when geneticists developed techniques for genetic work on bacteria, the common colon bacterium, *Escherichia coli*, became a favorite organism of genetic researchers (fig. 1.4). Because it had a generation time of only twenty minutes and only a small amount of genetic material, many research groups used it in their experiments. Still later, bacterial viruses, called *bacteriophages*, became very popular in genetics labs. The viruses are constructed of only a few types of protein molecules and a very small amount of genetic material. Some can replicate a hundredfold in an hour. Our point is not to list the major organisms geneticists use, but to suggest why they use some so commonly.



Figure 1.4 Scanning electron micrograph of *Escherichia coli* bacteria. These rod-shaped bacilli are magnified 18,000x.
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Comparative studies are usually done to determine which generalities discovered in the elite genetic organisms are really scientifically universal.

TECHNIQUES OF STUDY

Each area of genetics has its own particular techniques of study. Often the development of a new technique, or an improvement in a technique, has opened up major new avenues of research. As our technology has improved over the years, geneticists and other scientists have been able to explore at lower and lower levels of biological organization. Gregor Mendel, the father of genetics, did simple breeding studies of plants in a garden at his monastery in Austria in the middle of the nineteenth century. Today, with modern biochemical and biophysical techniques, it has become routine to determine the sequence of *nucleotides* (molecular subunits of DNA and RNA) that make up any particular gene. In fact, one of the most ambitious projects ever carried out in genetics is the mapping of the human genome, all 3.3 billion nucleotides that make up our genes. Only recently was the technology available to complete a project of this magnitude.

CLASSICAL, MOLECULAR, AND EVOLUTIONARY GENETICS

In the next three sections, we briefly outline the general subject areas covered in the book: classical, molecular, and evolutionary genetics.

Classical Genetics

Gregor Mendel discovered the basic rules of transmission genetics in 1866 by doing carefully controlled breeding experiments with the garden pea plant, *Pisum*

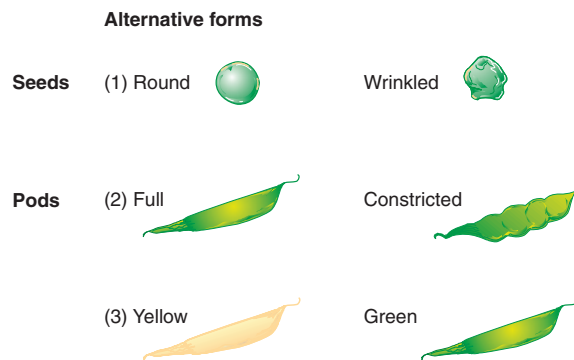


Figure 1.5 Mendel worked with garden pea plants. He observed seven traits of the plant—each with two discrete forms—that affected attributes of the seed, the pod, and the stem. For example, all plants had either round or wrinkled seeds, full or constricted pods, or yellow or green pods.

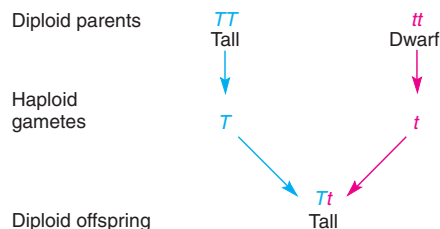


Figure 1.6 Mendel crossed tall and dwarf pea plants, demonstrating the rule of segregation. A diploid individual with two copies of the gene for tallness (T) per cell forms gametes that all have the T allele. Similarly, an individual that has two copies of the gene for shortness (t) forms gametes that all have the t allele. Cross-fertilization produces zygotes that have both the T and t alleles. When both forms are present (Tt), the plant is tall, indicating that the T allele is *dominant* to the *recessive* t allele.

sativum. He found that traits, such as pod color, were controlled by genetic elements that we now call *genes* (fig. 1.5). Alternative forms of a gene are called *alleles*. Mendel also discovered that adult organisms have two copies of each gene (*diploid* state); gametes receive just one of these copies (*haploid* state). In other words, one of the two parental copies segregates into any given gamete. Upon fertilization, the zygote gets one copy from each gamete, reconstituting the diploid number (fig. 1.6). When Mendel looked at the inheritance of several

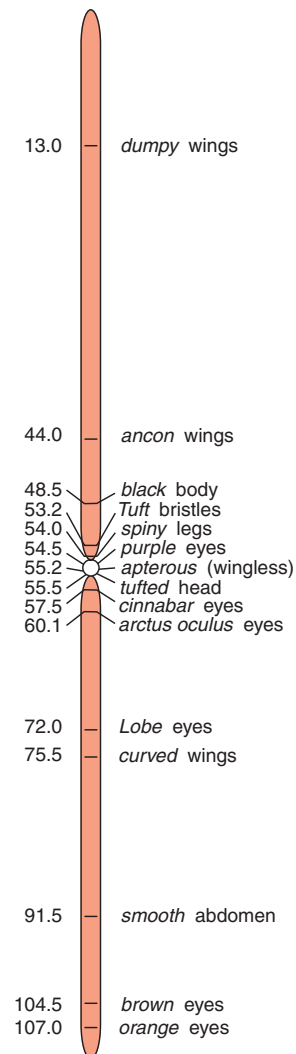


Figure 1.7 Genes are located in linear order on chromosomes, as seen in this diagram of chromosome 2 of *Drosophila melanogaster*, the common fruit fly. The centromere is a constriction in the chromosome. The numbers are map units.

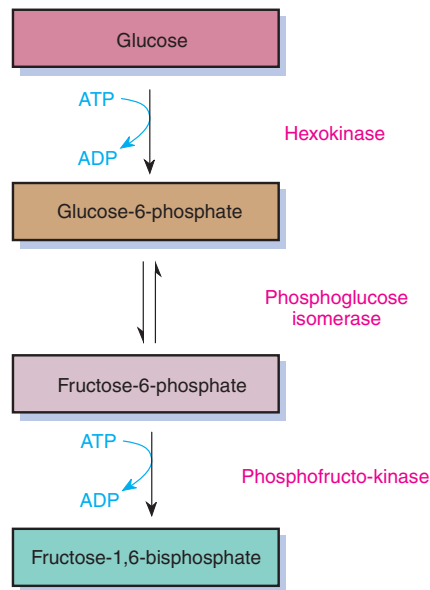


Figure 1.8 Biochemical pathways are the sequential changes that occur in compounds as cellular reactions modify them. In this case, we show the first few steps in the glycolytic pathway that converts glucose to energy. The pathway begins when glucose + ATP is converted to glucose-6-phosphate + ADP with the aid of the enzyme hexokinase. The enzymes are the products of genes.

traits at the same time, he found that they were inherited independently of each other. His work has been distilled into two rules, referred to as *segregation* and *independent assortment*. Scientists did not accept Mendel's work until they developed an understanding of the segregation of chromosomes during the latter half of the nineteenth century. At that time, in the year 1900, the science of genetics was born.

During much of the early part of this century, geneticists discovered many genes by looking for changed organisms, called *mutants*. Crosses were made to determine the genetic control of mutant traits. From this research evolved chromosomal mapping, the ability to locate the relative positions of genes on chromosomes by crossing certain organisms. The proportion of recombinant offspring, those with new combinations of parental alleles, gives a measure of the physical separation between genes on the same chromosomes in distances called *map units*. From this work arose the chromosomal theory of inheritance: Genes are located at fixed positions on chromosomes in a linear order (fig. 1.7, p. 9). This "beads on a string" model of gene

arrangement was not modified to any great extent until the middle of this century, after Watson and Crick worked out the structure of DNA.

In general, genes function by controlling the synthesis of proteins called *enzymes* that act as biological catalysts in biochemical pathways (fig. 1.8). G. Beadle and E. Tatum suggested that one gene controls the formation of one enzyme. Although we now know that many proteins are made up of subunits—the products of several genes—and that some genes code for proteins that are not enzymes and other genes do not code for proteins, the *one-gene-one-enzyme* rule of thumb serves as a general guideline to gene action.

Molecular Genetics

With the exception of some viruses, the genetic material of all cellular organisms is double-stranded DNA, a double helical molecule shaped like a twisted ladder. The backbones of the helices are repeating units of sugars (deoxyribose) and phosphate groups. The rungs of the

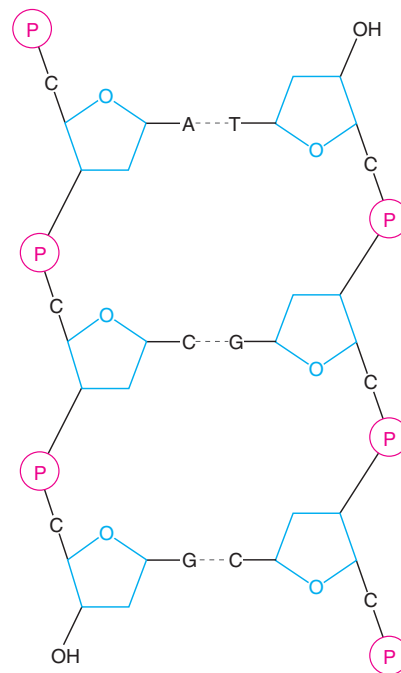


Figure 1.9 A look at a DNA double helix, showing the sugar-phosphate units that form the molecule's "backbone" and the base pairs that make up the "rungs." We abbreviate a phosphate group as a "P" within a circle; the pentagonal ring containing an oxygen atom is the sugar deoxyribose. Bases are either adenine, thymine, cytosine, or guanine (A, T, C, G).

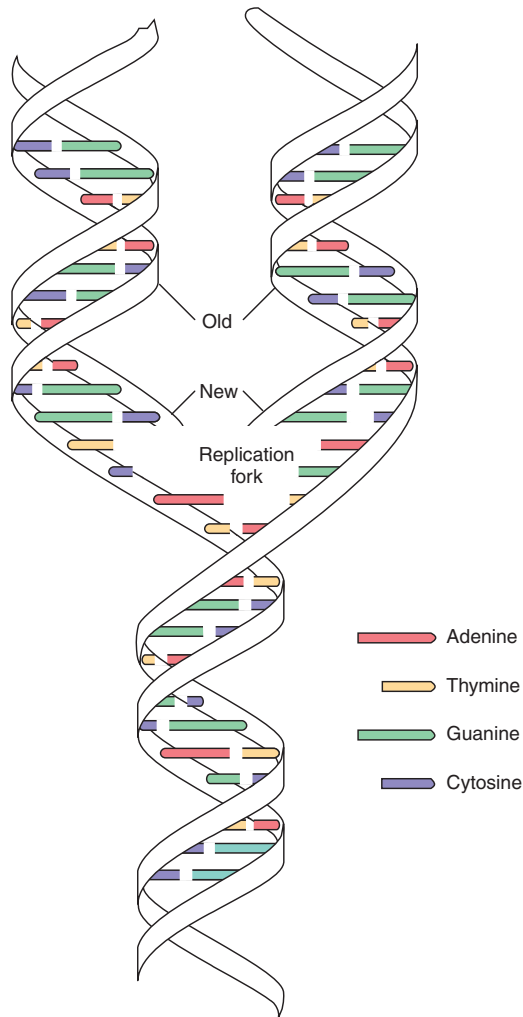


Figure 1.10 The DNA double helix unwinds during replication, and each half then acts as a template for a new double helix. Because of the rules of complementarity, each new double helix is identical to the original, and the two new double helices are identical to each other. Thus, an AT base pair in the original DNA double helix replicates into two AT base pairs, one in each of the daughter double helices.

ladder are base pairs, with one base extending from each backbone (fig. 1.9). Only four bases normally occur in DNA: adenine, thymine, guanine, and cytosine, abbreviated A, T, G, and C, respectively. There is no restriction on the order of bases on one strand. However, a relationship called *complementarity* exists between bases forming a rung. If one base of the pair is adenine, the other must be thymine; if one base is guanine, the other

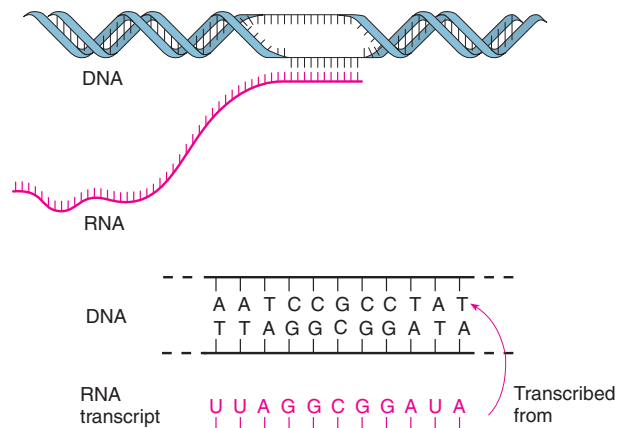


Figure 1.11 Transcription is the process that synthesizes RNA from a DNA template. Synthesis proceeds with the aid of the enzyme RNA polymerase. The DNA double helix partially unwinds during this process, allowing the base sequence of one strand to serve as a template for RNA synthesis. Synthesis follows the rules of DNA-RNA complementarity: A, T, G, and C of DNA pair with U, A, C, and G, respectively, in RNA. The resulting RNA base sequence is identical to the sequence that would form if the DNA were replicating instead, with the exception that RNA replaces thymine (T) with uracil (U).

must be cytosine. James Watson and Francis Crick deduced this structure in 1953, ushering in the era of molecular genetics.

The complementary nature of the base pairs of DNA made the mode of replication obvious to Watson and Crick: The double helix would “unzip,” and each strand would act as a template for a new strand, resulting in two double helices exactly like the first (fig. 1.10). Mutation, a change in one of the bases, could result from either an error in base pairing during replication or some damage to the DNA that was not repaired by the time of the next replication cycle.

Information is encoded in DNA in the sequence of bases on one strand of the double helix. During gene expression, that information is *transcribed* into RNA, the other form of nucleic acid, which actually takes part in protein synthesis. RNA differs from DNA in several respects: it has the sugar ribose in place of deoxyribose; it has the base uracil (U) in place of thymine (T); and it usually occurs in a single-stranded form. RNA is transcribed from DNA by the enzyme *RNA polymerase*, using DNA-RNA rules of complementarity: A, T, G, and C in DNA pair with U, A, C, and G, respectively, in RNA (fig. 1.11). The DNA information that is transcribed into RNA codes for the amino acid sequence of proteins. Three nucleotide bases form a *codon* that specifies one of the twenty

Table 1.2 The Genetic Code Dictionary of RNA

Codon	Amino Acid	Codon	Amino Acid	Codon	Amino Acid	Codon	Amino Acid
UUU	Phe	UCU	Ser	UAU	Tyr	UGU	Cys
UUC	Phe	UCC	Ser	UAC	Tyr	UGC	Cys
UUA	Leu	UCA	Ser	UAA	STOP	UGA	STOP
UUG	Leu	UCG	Ser	UAG	STOP	UGG	Trp
CUU	Leu	CCU	Pro	CAU	His	CGU	Arg
CUC	Leu	CCC	Pro	CAC	His	CGC	Arg
CUA	Leu	CCA	Pro	CAA	Gln	CGA	Arg
CUG	Leu	CCG	Pro	CAG	Gln	CGG	Arg
AUU	Ile	ACU	Thr	AAU	Asn	AGU	Ser
AUC	Ile	ACC	Thr	AAC	Asn	AGC	Ser
AUA	Ile	ACA	Thr	AAA	Lys	AGA	Arg
AUG	Met (START)	ACG	Thr	AAG	Lys	AGG	Arg
GUU	Val	GCU	Ala	GAU	Asp	GGU	Gly
GUC	Val	GCC	Ala	GAC	Asp	GGC	Gly
GUA	Val	GCA	Ala	GAA	Glu	GGA	Gly
GUG	Val	GCG	Ala	GAG	Glu	GGG	Gly

Note: A codon, specifying one amino acid, is three bases long (read in RNA bases in which U replaced the T of DNA). There are sixty-four different codons, specifying twenty naturally occurring amino acids (abbreviated by three letters: e.g., Phe is phenylalanine—see fig. 11.1 for the names and structures of the amino acids). Also present is *stop* (UAA, UAG, UGA) and *start* (AUG) information.

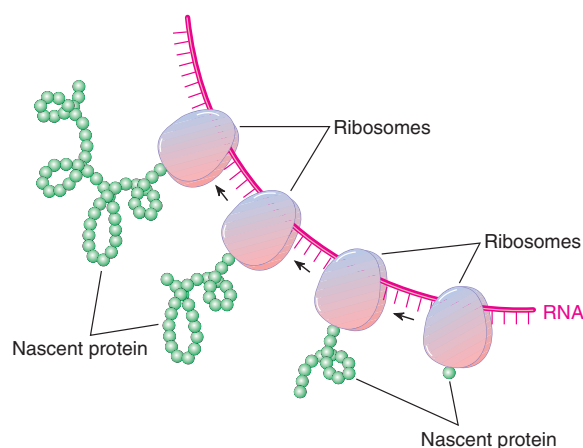


Figure 1.12 In prokaryotes, RNA translation begins shortly after RNA synthesis. A ribosome attaches to the RNA and begins reading the RNA codons. As the ribosome moves along the RNA, amino acids attach to the growing protein. When the process is finished, the completed protein is released from the ribosome, and the ribosome detaches from the RNA. As the first ribosome moves along, a second ribosome can attach at the beginning of the RNA, and so on, so that an RNA strand may have many ribosomes attached at one time.

naturally occurring amino acids used in protein synthesis. The sequence of bases making up the codons are referred to as the genetic code (table 1.2).

The process of *translation*, the decoding of nucleotide sequences into amino acid sequences, takes place at the ribosome, a structure found in all cells that is made up of RNA and proteins (fig. 1.12). As the RNA moves along the ribosome one codon at a time, one amino acid attaches to the growing protein for each codon.

The major control mechanisms of gene expression usually act at the transcriptional level. For transcription to take place, the RNA polymerase enzyme must be able to pass along the DNA; if this movement is prevented, transcription stops. Various proteins can bind to the DNA, thus preventing the RNA polymerase from continuing, providing a mechanism to control transcription. One particular mechanism, known as the *operon model*, provides the basis for a wide range of control mechanisms in prokaryotes and viruses. Eukaryotes generally contain no operons; although we know quite a bit about some control systems for eukaryotic gene expression, the general rules are not as simple.

In recent years, there has been an explosion of information resulting from *recombinant DNA techniques*. This revolution began with the discovery of *restriction endonucleases*, enzymes that cut DNA at specific se-

quences. Many of these enzymes leave single-stranded ends on the cut DNA. If a restriction enzyme acts on both a *plasmid*, a small, circular extrachromosomal unit found in some bacteria, and another piece of DNA (called foreign DNA), the two will be left with identical single-stranded free ends. If the cut plasmid and cut foreign DNA are mixed together, the free ends can re-form double helices, and the plasmid can take in a single piece of foreign DNA (fig. 1.13). Final repair processes create a completely closed circle of DNA. The hybrid plasmid is then reinserted into the bacterium. When the bacterium grows, it replicates the plasmid DNA, producing many copies of the foreign DNA. From that point, the foreign DNA can be isolated and sequenced, allowing researchers to determine the exact order of bases making up the foreign DNA. (In 2000, scientists announced the complete sequencing of the human genome.) That sequence can tell us much about how a gene works. In addition, the foreign genes can function within the bacterium, resulting in bacteria expressing the foreign genes and producing their protein products. Thus we have, for example, *E. coli* bacteria that produce human growth hormone.

This technology has tremendous implications in medicine, agriculture, and industry. It has provided the opportunity to locate and study disease-causing genes, such as the genes for cystic fibrosis and muscular dystrophy, as well as suggesting potential treatments. Crop plants and farm animals are being modified for better productivity by improving growth and disease resistance. Industries that apply the concepts of genetic engineering are flourishing.

One area of great interest to geneticists is cancer research. We have discovered that a single gene that has lost its normal control mechanisms (an *oncogene*) can cause changes that lead to cancer. These oncogenes exist normally in noncancerous cells, where they are called proto-oncogenes, and are also carried by viruses, where they are called viral oncogenes. Cancer-causing viruses are especially interesting because most of them are of the RNA type. AIDS is caused by one of these RNA viruses, which attacks one of the cells in the immune system. Cancer can also occur when genes that normally prevent cancer, genes called anti-oncogenes, lose function. Discovering the mechanism by which our immune system can produce millions of different protective proteins (*antibodies*) has been another success of modern molecular genetics.

Evolutionary Genetics

From a genetic standpoint, evolution is the change in allelic frequencies in a population over time. Charles Darwin described evolution as the result of natural selection. In the 1920s and 1930s, geneticists, primarily Sewall

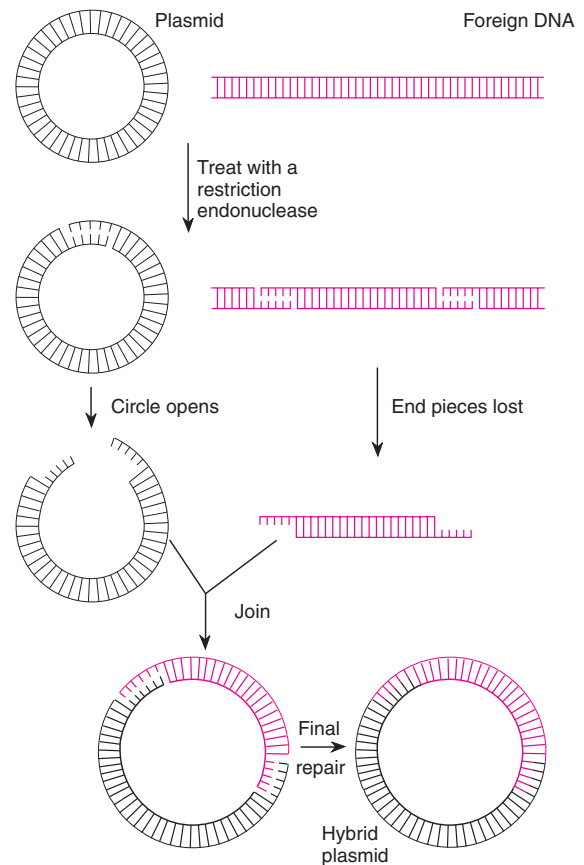


Figure 1.13 Hybrid DNA molecules can be constructed from a plasmid and a piece of foreign DNA. The ends are made compatible by cutting both DNAs with the same restriction endonuclease, leaving complementary ends. These ends will re-form double helices to form intact hybrid plasmids when the two types of DNA mix. A repair enzyme, DNA ligase, finishes patching the hybrid DNA within the plasmid. The hybrid plasmid is then reinjected into a bacterium, to be grown into billions of copies that will later be available for isolation and sequencing, or the hybrid plasmid can express the foreign DNA from within the host bacterium.

Wright, R. A. Fisher, and J. B. S. Haldane, provided algebraic models to describe evolutionary processes. The marriage of Darwinian theory and population genetics has been termed *neo-Darwinism*.

In 1908, G. H. Hardy and W. Weinberg discovered that a simple genetic equilibrium occurs in a population if the population is large, has random mating, and has negligible effects of mutation, migration, and natural selection. This equilibrium gives population geneticists a baseline for comparing populations to see if any evolutionary

processes are occurring. We can formulate a statement to describe the equilibrium condition: If the assumptions are met, the population will not experience changes in allelic frequencies, and these allelic frequencies will accurately predict the frequencies of *genotypes* (allelic combinations in individuals, e.g., *AA*, *Aa*, or *aa*) in the population.

Recently, several areas of evolutionary genetics have become controversial. Electrophoresis (a method for separating proteins and other molecules) and subsequent DNA sequencing have revealed that much more *polymorphism* (variation) exists within natural populations than older mathematical models could account for. One of the more interesting explanations for this variability is that it is neutral. That is, natural selection, the guiding force of evolution, does not act differentially on many, if not most, of the genetic differences found so commonly in nature. At first, this theory was quite controversial, attracting few followers. Now it seems to be the view the

majority accept to explain the abundance of molecular variation found in natural populations.

Another controversial theory concerns the rate of evolutionary change. It is suggested that most evolutionary change is not gradual, as the fossil record seems to indicate, but occurs in short, rapid bursts, followed by long periods of very little change. This theory is called *punctuated equilibrium*.

A final area of evolutionary biology that has generated much controversy is the theory of *sociobiology*. Sociobiologists suggest that social behavior is under genetic control and is acted upon by natural selection, as is any morphological or physiological trait. This idea is controversial mainly as it applies to human beings; it calls altruism into question and suggests that to some extent we are genetically programmed to act in certain ways. People have criticized the theory because they feel it justifies racism and sexism.

S U M M A R Y

The purpose of this chapter has been to provide a brief history of genetics and a brief overview of the following twenty chapters. We hope it serves to introduce the material and to provide a basis for early synthesis of some of the material that, of necessity, is presented in the discrete units called chapters. This chapter also differs from all the others because it lacks some of the end materials that

should be of value to you as you proceed: solved problems, and exercises and problems. These features are presented chapter by chapter throughout the remainder of the book. At the end of the book, we provide answers to exercises and problems and a glossary of all **boldface** words throughout the book.

Suggested Readings for chapter 1 are on page B-1.

2

MENDEL'S PRINCIPLES

STUDY OBJECTIVES

1. To understand that genes are discrete units that control the appearance of an organism 17
2. To understand Mendel's rules of inheritance: segregation and independent assortment 18
3. To understand that dominance is a function of the interaction of alleles; similarly, epistasis is a function of the interaction of nonallelic genes 22
4. To define how genes generally control the production of enzymes and thus the fate of biochemical pathways 37

STUDY OUTLINE

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Segregation 18

Rule of Segregation 18

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Box 2.2 Did Mendel Cheat? 30



The garden pea plant, *Pisum sativum*.

(© Adam Hart-Davis/SPL/Photo Researchers, Inc.)

Genetics is concerned with the transmission, expression, and evolution of genes, the molecules that control the function, development, and ultimate appearance of individuals. In this section of the book, we will look at the rules of transmission that govern genes and affect their passage from one generation to the next. Gregor Johann Mendel discovered these rules of inheritance; we derive and expand upon his rules in this chapter (fig. 2.1).

In 1900, three botanists, Carl Correns of Germany, Erich von Tschermak of Austria, and Hugo de Vries of Holland, defined the rules governing the transmission of traits from parent to offspring. Some historical controversy exists as to whether these botanists actually rediscovered Mendel's rules by their own research or whether their research led them to Mendel's original paper. In any case, all three made important contributions to the early stages of genetics. The rules had been published previously, in 1866, by an obscure Austrian monk, Gregor Johann Mendel. Although his work was widely available after 1866, the scientific community was not ready to appreciate Mendel's great contribution until the turn of the century. There are at least four reasons for this lapse of thirty-four years.



Figure 2.1 Gregor Johann Mendel (1822–84). (Reproduced by permission of the Moravski Museum, Mendelianum.)

First, before Mendel's experiments, biologists were primarily concerned with explaining the transmission of characteristics that could be measured on a continuous scale, such as height, cranium size, and longevity. They were looking for rules of inheritance that would explain such **continuous variations**, especially after Darwin put forth his theory of evolution in 1859 (see chapter 21). Mendel, however, suggested that inherited characteristics were discrete and constant (**discontinuous**): peas, for example, were either yellow or green. Thus, evolutionists were looking for small changes in traits with continuous variation, whereas Mendel presented them with rules for discontinuous variation. His principles did not seem to apply to the type of variation that biologists thought prevailed. Second, there was no physical element identified with Mendel's inherited entities. One could not say, upon reading Mendel's work, that a certain subunit of the cell followed Mendel's rules. Third, Mendel worked with large numbers of offspring and converted these numbers to ratios. Biologists, practitioners of a very descriptive science at the time, were not well trained in mathematical tools. And last, Mendel was not well known and did not persevere in his attempts to convince the academic community that his findings were important.

Between 1866 and 1900, two major changes took place in biological science. First, by the turn of the century, not only had scientists discovered chromosomes, but they also had learned to understand chromosomal movement during cell division. Second, biologists were better prepared to handle mathematics by the turn of the century than they were during Mendel's time.

MENDEL'S EXPERIMENTS

Gregor Mendel was an Austrian monk (of Brünn, Austria, which is now Brno, Czech Republic). In his experiments, he tried to **crossbreed** plants that had discrete, nonoverlapping characteristics and then to observe the distribution of these characteristics over the next several generations. Mendel worked with the common garden pea plant, *Pisum sativum*. He chose the pea plant for at least three reasons: (1) The garden pea was easy to cultivate and had a relatively short life cycle. (2) The plant had discontinuous characteristics such as flower color and pea texture. (3) In part because of its anatomy, pollination of the plant was easy to control. Foreign pollen could be kept out, and **cross-fertilization** could be accomplished artificially.

Figure 2.2 shows a cross section of the pea flower that indicates the keel, in which the male and female parts develop. Normally, **self-fertilization** occurs when pollen falls onto the stigma before the bud opens. Mendel cross-fertilized the plants by opening the keel of

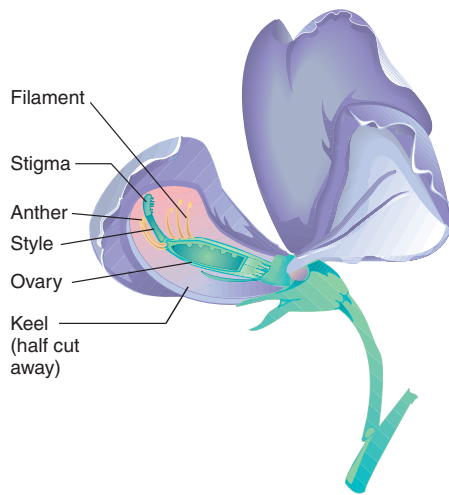


Figure 2.2 Anatomy of the garden pea plant flower. The female part, the pistil, is composed of the stigma, its supporting style, and the ovary. The male part, the stamen, is composed of the pollen-producing anther and its supporting filament.

a flower before the anthers matured and placing pollen from another plant on the stigma. In the more than ten thousand plants Mendel examined, only a few were fertilized other than the way he had intended (either self- or cross-pollinated).

Mendel used plants obtained from suppliers and grew them for two years to ascertain that they were homogeneous, or true-breeding, for the particular characteristic under study. He chose for study the seven characteristics shown in figure 2.3. Take as an example the characteristic of plant height. Although height is often continuously distributed, Mendel used plants that displayed only two alternatives: tall or dwarf. He made the crosses shown in figure 2.4. In the parental, or P_1 , generation, dwarf plants pollinated tall plants, and, in a **reciprocal cross**, tall plants pollinated dwarf plants, to determine whether the results were independent of the parents' sex. As we will see later on, some traits follow inheritance patterns related to the sex of the parent carrying the traits. In those cases, reciprocal crosses give different results; with Mendel's tall and dwarf pea plants, the results were the same.

Offspring of the cross of P_1 individuals are referred to as the first **filial generation**, or F_1 . Mendel also referred to them as **hybrids** because they were the offspring of unlike parents (tall and dwarf). We will specifically refer to the offspring of tall and dwarf peas as **monohybrids** because they are hybrid for only one characteristic (height). Since all the F_1 offspring plants were tall, Mendel referred to tallness as the **dominant** trait. The alternative, dwarfness, he referred to as **recessive**. Differ-

ent forms of a gene that exist within a population are termed **alleles**. The terms *dominant* and *recessive* are used to describe both the relationship between the alleles and the traits they control. Thus, we say that both the allele for tallness and the trait, tall, are dominant. Dominance applies to the appearance of the trait when both a dominant and a recessive allele are present. It does not imply that the dominant trait is better, is more abundant, or will increase over time in a population.

When the F_1 offspring of figure 2.4 were self-fertilized to produce the F_2 generation, both tall and dwarf offspring occurred; the dwarf characteristic reappeared. Among the F_2 offspring, Mendel observed 787 tall and 277 dwarf plants for a ratio of 2.84:1. It is an indication of Mendel's insight that he recognized in these numbers an approximation to a 3:1 ratio, a ratio that suggested to him the mechanism of inheritance at work in pea plant height.

SEGREGATION

Rule of Segregation

Mendel assumed that each plant contained two determinants (which we now call **genes**) for the characteristic of height. For example, a hybrid F_1 pea plant possesses the dominant allele for tallness and the recessive allele for dwarfness for the gene that determines plant height. A pair of alleles for dwarfness is required to develop the recessive phenotype. Only one of these alleles is passed into a single **gamete**, and the union of two gametes to form a zygote restores the double complement of alleles. The fact that the recessive trait reappears in the F_2 generation shows that the allele controlling it was hidden in the F_1 individual and passed on unaffected. This explanation of the passage of discrete trait determinants, or genes, comprises Mendel's first principle, the **rule of segregation**. The rule of segregation can be summarized as follows: A gamete receives only one allele from the pair of alleles an organism possesses; fertilization (the union of two gametes) reestablishes the double number. We can visualize this process by redrawing figure 2.4 using letters to denote the alleles. Mendel used capital letters to denote alleles that control dominant traits and lowercase letters for alleles that control recessive traits. Following this notation, T refers to the allele controlling tallness and t refers to the allele controlling shortness (dwarf stature). From figure 2.5, we can see that Mendel's rule of segregation explains the homogeneity of the F_1 generation (all tall) and the 3:1 ratio of tall-to-dwarf offspring in the F_2 generation.

Let us define some terms. The **genotype** of an organism is the gene combination it possesses. In figure 2.5,

Alternative forms

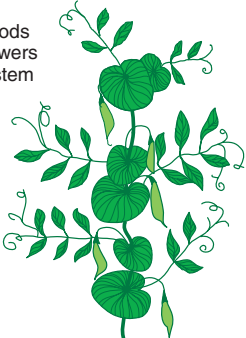
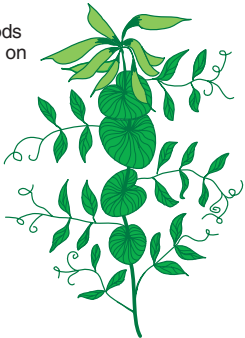
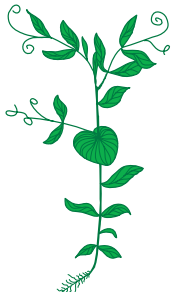
Seeds	(1) Round		Wrinkled	
	(2) Yellow cotyledons		Green cotyledons	
	(3) Gray coat (violet flowers)		White coat (white flowers)	
Pods	(4) Full		Constricted	
	(5) Green		Yellow	
Stem	(6) Axial pods and flowers along stem		Terminal pods and flowers on top of stem	
	(7) Tall (6–7 ft)		Dwarf (3/4–1 ft)	

Figure 2.3 Seven characteristics that Mendel observed in peas. Traits in the left column are dominant.

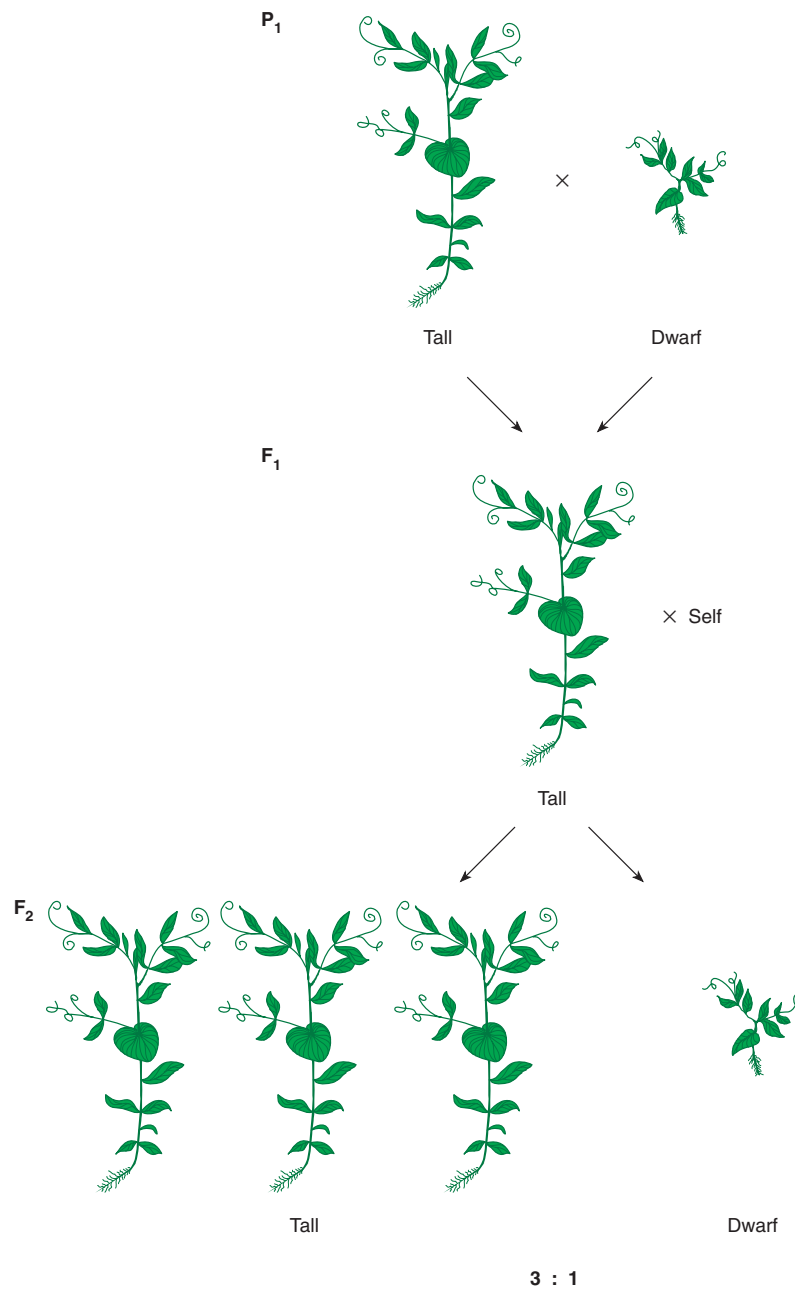


Figure 2.4 First two offspring generations from the cross of tall plants with dwarf plants.

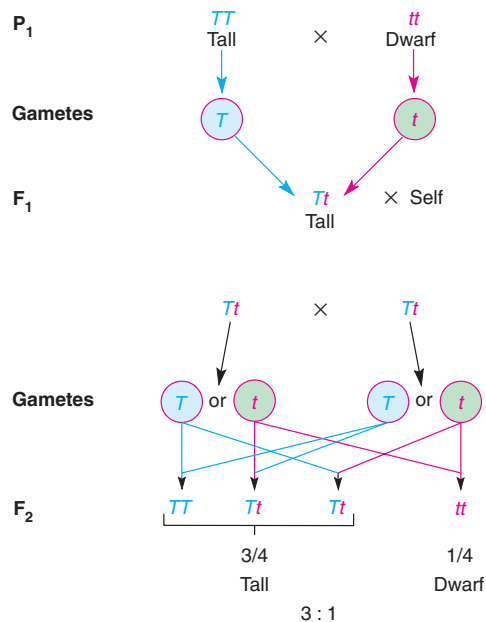


Figure 2.5 Assigning genotypes to the cross in figure 2.4.

the genotype of the parental tall plant is TT ; that of the F_1 tall plant is Tt . **Phenotype** refers to the observable attributes of an organism. Plants with either of the two genotypes TT or Tt are phenotypically tall. Genotypes come in two general classes: **homozygotes**, in which both alleles are the same, as in TT or tt , and **heterozygotes**, in which the two alleles are different, as in Tt . William Bateson coined these last two terms in 1901. Danish botanist Wilhelm Johannsen first used the word *gene* in 1909.

If we look at figure 2.5, we can see that the TT homozygote can produce only one type of gamete, the T -bearing kind, and the tt homozygote can similarly produce only t -bearing gametes. Thus, the F_1 individuals are uniformly heterozygous Tt , and each F_1 individual can produce two kinds of gametes in equal frequencies, T - or t -bearing. In the F_2 generation, these two types of gametes randomly pair during fertilization. Figure 2.6 shows three ways of picturing this process.

Testing the Rule of Segregation

We can see from figure 2.6 that the F_2 generation has a phenotypic ratio of 3:1, the classic Mendelian ratio. However, we also see a genotypic ratio of 1:2:1 for dominant homozygote:heterozygote:recessive homozygote. Demonstrating this genotypic ratio provides a good test of Mendel's rule of segregation.

The simplest way to test the hypothesis is by **progeny testing**, that is, by self-fertilizing F_2 individuals to

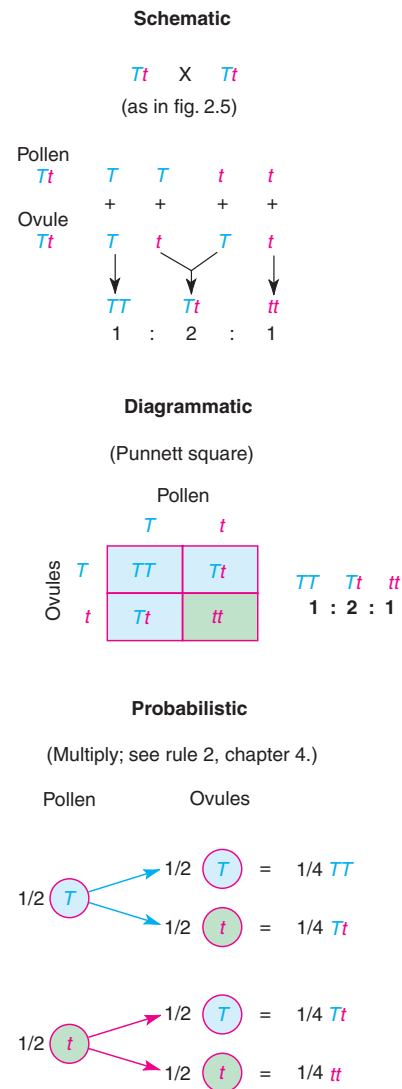


Figure 2.6 Methods of determining F_2 genotypic combinations in a self-fertilized monohybrid. The Punnett square diagram is named after the geneticist Reginald C. Punnett.

produce an F_3 generation, which Mendel did (fig. 2.7). Treating the rule of segregation as a hypothesis, it is possible to predict the frequencies of the phenotypic classes that would result. The dwarf F_2 plants should be recessive homozygotes, and so, when **selfed** (self-fertilized), they should produce only t -bearing gametes and only dwarf offspring in the F_3 generation. The tall F_2 plants, however, should be a heterogeneous group, one-third of which should be homozygous TT and two-thirds heterozygous Tt . The tall homozygotes, when selfed, should produce only tall F_3 offspring (genotypically TT). However, the F_2 heterozygotes, when selfed, should produce

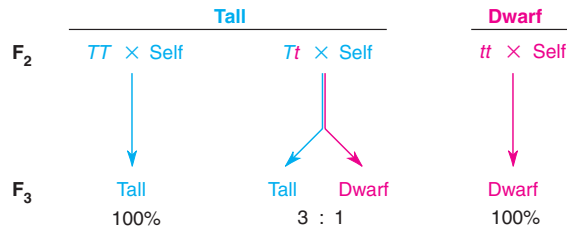


Figure 2.7 Mendel self-fertilized F₂ tall and dwarf plants. He found that all the dwarf plants produced only dwarf progeny. Among the tall plants, 72% produced both tall and dwarf progeny in a 3:1 ratio.

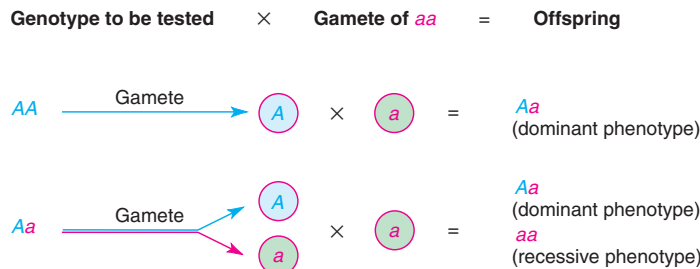


Figure 2.8 Testcross. In a testcross, the phenotype of an offspring is determined by the allele the offspring inherits from the parent with the genotype being tested.

tall and dwarf offspring in a ratio identical to that the selfed F_1 plants produced: three tall to one dwarf offspring. Mendel found that all the dwarf (homozygous) F_2 plants bred true as predicted. Among the tall, 28% (28/100) bred true (produced only tall offspring) and 72% (72/100) produced both tall and dwarf offspring. Since the prediction was one-third (33.3%) and two-thirds (66.7%), respectively, Mendel's observed values were very close to those predicted. We thus conclude that Mendel's progeny-testing experiment confirmed his hypothesis of segregation. In fact, a statistical test—developed in chapter 4—would also support this conclusion.

Another way to test the segregation rule is to use the extremely useful method of the **testcross**, that is, a cross of any organism with a recessive homozygote. (Another type of cross, a **backcross**, is the cross of a progeny with

an individual that has a parental genotype. Hence, a testcross can often be a backcross.) Since the gametes of the recessive homozygote contain only recessive alleles, the alleles that the gametes of the other parent carry will determine the phenotypes of the offspring. If a gamete from the organism being tested contains a recessive allele, the resulting F_1 organism will have a recessive phenotype; if it contains a dominant allele, the F_1 organism will have a dominant phenotype. Thus, in a testcross, the genotypes of the gametes from the organism being tested determine the phenotypes of the offspring (fig. 2.8). A testcross of the tall F_2 plants in figure 2.5 would produce the results shown in figure 2.9. These results further confirm Mendel's rule of segregation.

DOMINANCE IS NOT UNIVERSAL

If dominance were universal, the heterozygote would always have the same phenotype as the dominant homozygote, and we would always see the 3:1 ratio when heterozygotes are crossed. If, however, the heterozygote were distinctly different from both homozygotes, we

Tall (two classes)
 $TT \times tt = \text{all } Tt$
 $Tt \times tt = Tt : tt$
1 : 1

Figure 2.9 Testcrossing the dominant phenotype of the F_2 generation from figure 2.5.

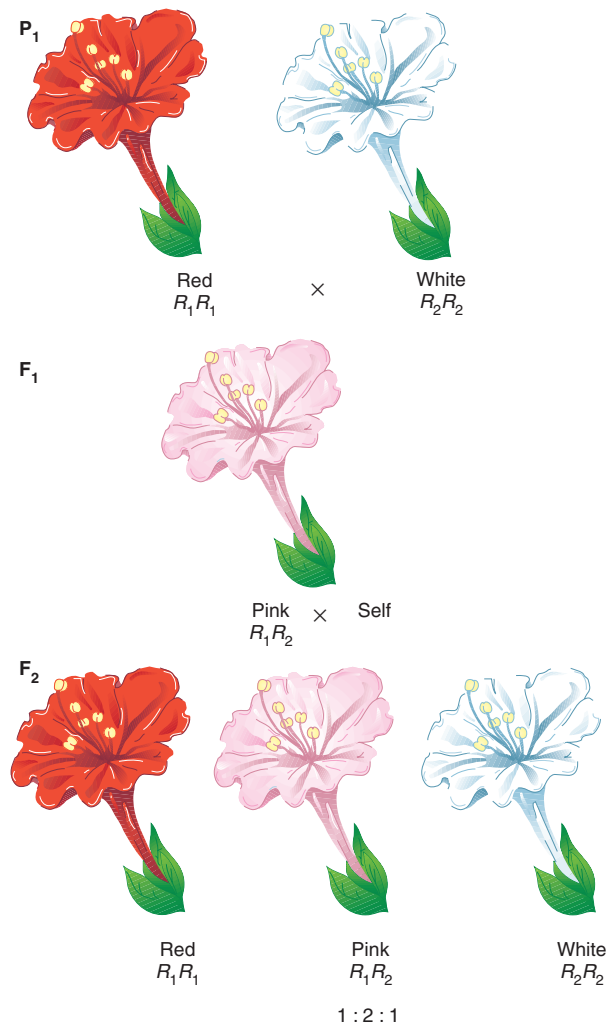


Figure 2.10 Flower color inheritance in the four-o'clock plant: an example of partial, or incomplete, dominance.

would see a 1:2:1 ratio of phenotypes when heterozygotes are crossed. In **partial dominance** (or **incomplete dominance**), the phenotype of the heterozygote falls between those of the two homozygotes. An example occurs in flower petal color in some plants.

Using four-o'clock plants (*Mirabilis jalapa*), we can cross a plant that has red flower petals with another that has white flower petals; the offspring will have pink flower petals. If these pink-flowered F₁ plants are crossed, the F₂ plants appear in a ratio of 1:2:1, having red, pink, or white flower petals, respectively (fig. 2.10). The pink-flowered plants are heterozygotes that have a petal color intermediate between the red and white colors of the homozygotes. In this case, one allele (R_1) specifies red pigment color, and another allele specifies no color (R_2 ; the flower petals have a white background

color). Flowers in heterozygotes (R_1R_2) have about half the red pigment of the flowers in red homozygotes (R_1R_1) because the heterozygotes have only one copy of the allele that produces color, whereas the homozygotes have two copies.

As technology has improved, we have found more and more cases in which we can differentiate the heterozygote. It is now clear that dominance and recessiveness are phenomena dependent on which alleles are interacting and on what phenotypic level we are studying. For example, in Tay-Sachs disease, homozygous recessive children usually die before the age of three after suffering severe nervous system degeneration; heterozygotes seem to be normal. As biologists have discovered how the disease works, they have made the detection of the heterozygotes possible.

As with many genetic diseases, the culprit is a defective **enzyme** (protein catalyst). Afflicted homozygotes have no enzyme activity, heterozygotes have about half the normal level, and, of course, homozygous normal individuals have the full level. In the case of Tay-Sachs disease, the defective enzyme is hexosaminidase-A, needed for proper lipid metabolism. Modern techniques allow technicians to assay the blood for this enzyme and to identify heterozygotes by their intermediate level of enzyme activity. Two heterozygotes can now know that there is a 25% chance that any child they bear will have the disease. They can make an educated decision as to whether or not to have children.

The other category in which the heterozygote is discernible occurs when the heterozygous phenotype is not on a scale somewhere between the two homozygotes, but actually expresses both phenotypes simultaneously. We refer to this situation as **codominance**. For example, people with blood type AB are heterozygotes who express both the A and B alleles for blood type (see the section entitled "Multiple Alleles" for more information about blood types). Electrophoresis (a technique described in chapter 5) lets us see proteins directly and also gives us many examples of codominance when we can see the protein products of both alleles.

NOMENCLATURE

Throughout the last century, botanists, zoologists, and microbiologists have adopted different methods for naming alleles. Botanists and mammalian geneticists tend to prefer the capital-lowercase scheme. *Drosophila* geneticists and microbiologists have adopted schemes that relate to the **wild-type**. The wild-type is the phenotype of the organism commonly found in nature. Though other naturally occurring phenotypes of the same species may also be present, there is usually an agreed-upon common

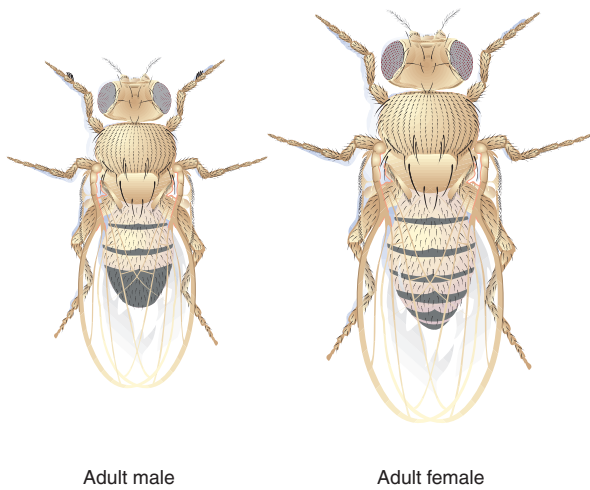


Figure 2.11 Wild-type fruit fly, *Drosophila melanogaster*.

phenotype that is referred to as the wild-type. For fruit flies (*Drosophila*), organisms commonly used in genetic studies, the wild-type has red eyes and round wings (fig. 2.11). Alternatives to the wild-type are referred to as **mutants** (fig. 2.12). Thus, red eyes are wild-type, and white eyes are mutant. Fruit fly genes are named after the mutant, beginning with a capital letter if the mutation is dominant and a lowercase letter if it is recessive. Table 2.1 gives some examples. The wild-type allele often carries the symbol of the mutant with a + added as a

Table 2.1 Some Mutants of *Drosophila*

Mutant Designation	Description	Dominance Relationship to Wild-Type
abrupt (<i>ab</i>)	Shortened, longitudinal, median wing vein	Recessive
amber (<i>amb</i>)	Pale yellow body	Recessive
black (<i>b</i>)	Black body	Recessive
Bar (<i>B</i>)	Narrow, vertical eye	Dominant
dumpy (<i>dp</i>)	Reduced wings	Recessive
Hairless (<i>H</i>)	Various bristles absent	Dominant
white (<i>w</i>)	White eye	Recessive
white-apricot (<i>w^a</i>)	Apricot-colored eye (allele of white eye)	Recessive

superscript; by definition, every mutant has a wild-type allele as an alternative. For example, *w* stands for the white-eye allele, a recessive mutation. The wild-type (red eyes) is thus assigned the symbol *w⁺*. Hairless is a dominant allele with the symbol *H*. Its wild-type allele is denoted as *H⁺*. Sometimes geneticists use the + symbol alone for the wild-type, but only when there will be no confusion about its use. If we are discussing eye color only, then + is clearly the same as *w⁺*: both mean red eyes. However, if we are discussing both eye color and bristle morphology, the + alone could refer to either of the two aspects of the phenotype and should be avoided.

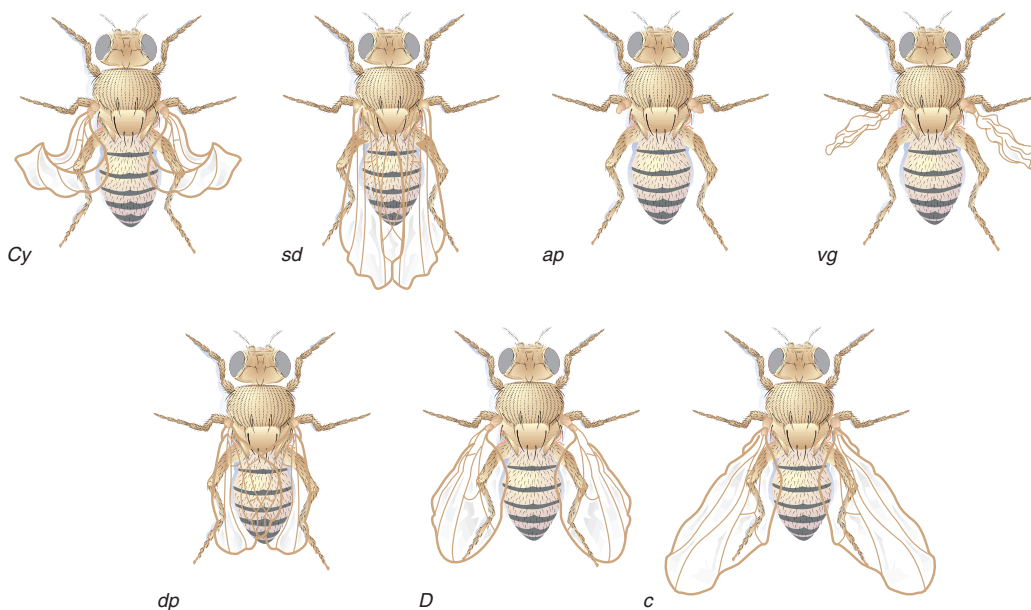


Figure 2.12 Wing mutants of *Drosophila melanogaster* and their allelic designations: *Cy*, curly; *sd*, scalloped; *ap*, apterous; *vg*, vestigial; *dp*, dumpy; *D*, Dichaete; *c*, curved.

Table 2.2 ABO Blood Types with Immunity Reactions

Blood Type Corresponding to Antigens on Red Blood Cells	Antibodies in Serum	Genotype	Reaction of Red Cells to Anti-A Antibodies	Reaction of Red Cells to Anti-B Antibodies
O	Anti-A and anti-B	<i>ii</i>	—	—
A	Anti-B	$I^A I^A$ or $I^A i$	+	—
B	Anti-A	$I^B I^B$ or $I^B i$	—	+
AB	None	$I^A I^B$	+	+

MULTIPLE ALLELES

A given gene can have more than two alleles. Although any particular individual can have only two, many alleles of a given gene may exist in a population. The classic example of multiple human alleles is in the ABO blood group, which Karl Landsteiner discovered in 1900. This is the best known of all the red-cell antigen systems primarily because of its importance in blood transfusions. There are four blood-type phenotypes produced by three alleles (table 2.2). The I^A and I^B alleles are responsible for the production of the A and B antigens found on the surface of the erythrocytes (red blood cells). Antigens are substances, normally foreign to the body, that induce the immune system to produce antibodies (proteins that bind to the antigens). The ABO system is unusual because antibodies can be present (e.g., anti-B antibodies can exist in a type A person) without prior exposure to the antigen. Thus, people with a particular ABO antigen on their red cells will have in their serum the antibody against the

other antigen: type A persons have A antigen on their red cells and anti-B antibody in their serum; type B persons have B antigen on their red cells and anti-A antibody in their serum; type O persons do not have either antigen but have both antibodies in their serum; and type AB persons have both A and B antigens and form neither anti-A or anti-B antibodies in their serum.

The I^A and I^B alleles, coding for glycosyl transferase enzymes, each cause a different modification to the terminal sugars of a mucopolysaccharide (H structure) found on the surface of red blood cells (fig. 2.13). They are codominant because both modifications (antigens) are present in a heterozygote. In fact, whichever enzyme (product of the I^A or I^B allele) reaches the H structure first will modify it. Once modified, the H structure will not respond to the other enzyme. Therefore, both A and B antigens will be produced in the heterozygote in roughly equal proportions. The i allele causes no change to the H structure: because of a mutation it produces a nonfunctioning enzyme. The i allele and its phenotype are recessive; the presence of the I^A or I^B allele, or both,

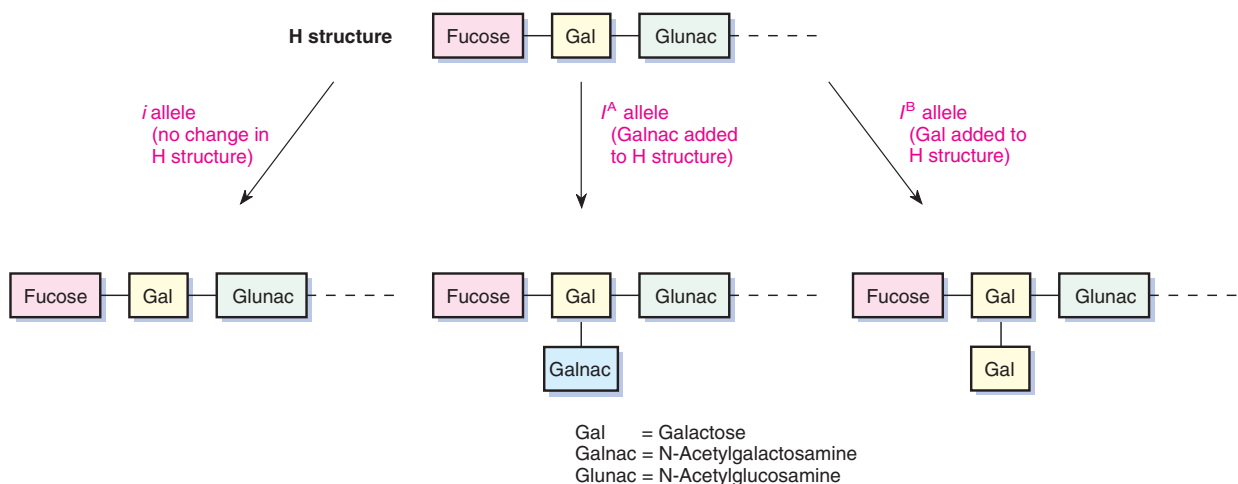


Figure 2.13 Function of the I^A , I^B , and i alleles of the ABO gene. The gene products of the I^A and I^B alleles of the ABO gene affect the terminal sugars of a mucopolysaccharide (H structure) found on red blood cells. The gene products of the I^A and I^B alleles are the enzymes alpha-3-N-acetyl-D-galactosaminyltransferase and alpha-3-D-galactosyltransferase, respectively.

will modify the H product, thus masking the fact that the *i* allele was ever there.

Adverse reactions to blood transfusions primarily occur because the antibodies in the recipient's serum react with the antigens on the donor's red blood cells. Thus, type A persons cannot donate blood to type B persons. Type B persons have anti-A antibody, which reacts with the A antigen on the donor red cells and causes the cells to clump.

Since both I^A and I^B are dominant to the *i* allele, this system not only shows multiple allelism, it also demonstrates both codominance and simple dominance. (As with virtually any system, intense study yields more information, and subgroups of type A are known. We will not, however, deal with that complexity here.) According to the American Red Cross, 46% of blood donors in the United States are type O, 40% are type A, 10% are type B, and 4% are type AB.

Many other genes also have multiple alleles. In some plants, such as red clover, there is a gene, the *S* gene, with several hundred alleles that prevent self-fertilization. This means that a pollen grain is not capable of forming a successful pollen tube in the style if the pollen grain or its parent plant has a self-incompatibility allele that is also present in the plant to be fertilized. Thus, pollen grains from a flower falling on its own stigma are rejected. Only a pollen grain with either a different self-incompatibility allele or from a parent plant with different self-incompatibility alleles is capable of fertilization; this avoids inbreeding. Thus, over evolutionary time, there has been selection for many alleles of this gene. Presumably, a foreign plant would not want to be mistaken for the same plant, providing the selective pressure for many alleles to survive in a population. Recent research has indicated that the products of the *S* alleles are ribonuclease enzymes, enzymes that destroy RNA. Researchers are interested in discovering the molecular mechanisms for this pollen rejection.

In *Drosophila*, numerous alleles of the white-eye gene exist, and people have numerous hemoglobin alleles. In fact, multiple alleles are the rule rather than the exception.

INDEPENDENT ASSORTMENT

Mendel also analyzed the inheritance pattern of traits observed two at a time. He looked, for instance, at plants that differed in the form and color of their peas: he crossed true-breeding (homozygous) plants that had seeds that were round and yellow with plants that produced seeds that were wrinkled and green. Mendel's results appear in figure 2.14. The F_1 plants all had round, yellow seeds, which demonstrated that round was dominant to wrinkled and yellow was dominant to green. When these F_1 plants were self-fertilized, they produced

an F_2 generation that had all four possible combinations of the two seed characteristics: round, yellow seeds; round, green seeds; wrinkled, yellow seeds; and wrinkled, green seeds. The numbers Mendel reported in these categories were 315, 108, 101, and 32, respectively. Dividing each number by 32 gives a 9.84 to 3.38 to 3.16 to 1.00 ratio, which is very close to a 9:3:3:1 ratio. As you will see, this is the ratio we would expect if the genes governing these two traits behaved independently of each other.

In figure 2.14, the letter *R* is assigned to the dominant allele, round, and *r* to the recessive allele, wrinkled; *Y* and *y* are used for yellow and green color, respectively. In figure 2.15, we have rediagrammed the cross in figure 2.14. The P_1 plants in this cross produce only one type of gamete each, *RY* for the parent with the dominant traits and *ry* for the parent with the recessive traits. The resulting F_1 plants are heterozygous for both genes (**dihybrid**). Self-fertilizing the dihybrid (*RrYy*) produces the F_2 generation.

In constructing the **Punnett square** in figure 2.15 to diagram the F_2 generation, we make a critical assumption: The four types of gametes from each parent will be produced in equal numbers, and hence every offspring category, or "box," in the square is equally likely. Thus, because sixteen boxes make up the Punnett square (named after its inventor, Reginald C. Punnett), the ratio of F_2 offspring should be in sixteenths. Grouping the F_2 offspring by phenotype, we find there are 9/16 that have round, yellow seeds; 3/16 that have round, green seeds; 3/16 that have wrinkled, yellow seeds; and 1/16 that have wrinkled, green seeds. This is the origin of the expected 9:3:3:1 F_2 ratio.



Reginald C. Punnett (1875–1967).
From *Genetics*, 58 (1968): frontispiece.
Courtesy of the Genetics Society of
America.

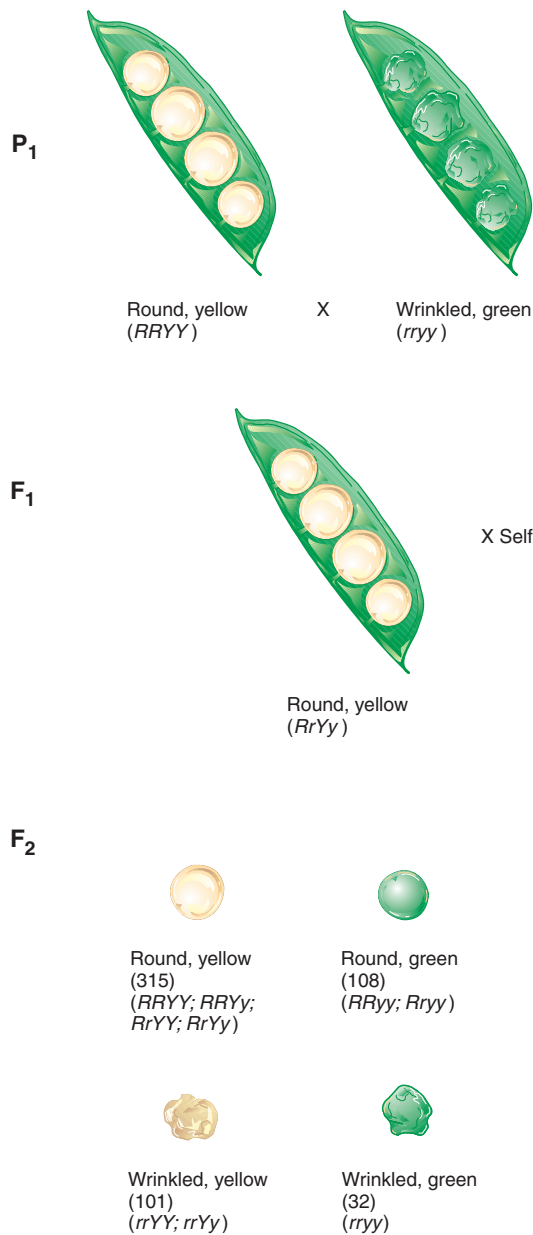


Figure 2.14 Independent assortment in garden peas.

Rule of Independent Assortment

This ratio comes about because the two characteristics behave independently. The F₁ plants produce four types of gametes (check fig. 2.15): *RY*, *Ry*, *rY*, and *ry*. These gametes occur in equal frequencies. Regardless of which seed shape allele a gamete ends up with, it has a 50:50

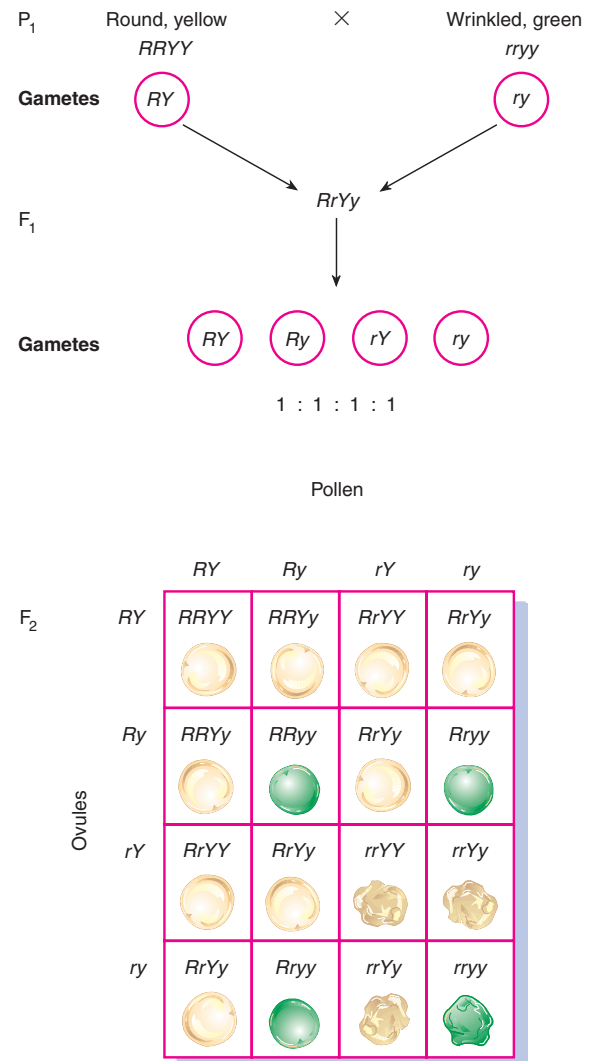


Figure 2.15 Assigning genotypes to the cross in figure 2.14.

chance of getting either of the alleles for color—the two genes are segregating, or assorting, independently. This is the essence of Mendel's second rule, the **rule of independent assortment**, which states that alleles for one gene can segregate independently of alleles for other genes. Are the alleles for the two characteristics of color and form segregating properly according to Mendel's first principle?

If we look only at seed shape (see fig. 2.14), we find that a homozygote with round seeds was crossed with a homozygote with wrinkled seeds in the P₁ generation (*RR* × *rr*). This cross yields only heterozygous plants with round seeds (*Rr*) in the F₁ generation. When these

BOX 2.1

In February and March of 1865, Mendel delivered two lectures to the Natural History Society of Brünn. These were published as a single forty-eight-page article handwritten in German. The article appeared in the 1865 *Proceedings of the Society*, which came out in 1866. It was entitled “*Versuche über Pflanzen-Hybriden*,” which means “Experiments in Plant Hybridization.” Following are some paragraphs from the English translation to give us some sense of the original.

In his introductory remarks, Mendel writes:

That, so far, no generally applicable law governing the formation and development of hybrids has been successfully formulated can hardly be wondered at by anyone who is acquainted with the extent of the task, and can appreciate the difficulties with which experiments of this class have to contend. A final decision can only be arrived at when we shall have before us the results of detailed experiments made on plants belonging to the most diverse orders.

Those who survey the work done in this department will arrive at the conviction that among all the numerous experiments made, not one has been carried out to such an extent and in such a way as to make it possible to determine the number of different forms under which the offspring of hybrids appear, or to arrange these forms with certainty according to their separate genera-

tions, or definitely to ascertain their statistical relations. . . .

The paper now presented records the results of such a detailed experiment. This experiment was practically confined to a small plant group, and is now, after eight years' pursuit, concluded in all essentials. Whether the plan upon which the separate experiments were conducted and carried out was the best suited to attain the desired end is left to the friendly decision of the reader.

After discussing the origin of his seeds and the nature of the experiments, Mendel discusses the F_1 , or hybrid, generation:

This is precisely the case with the Pea hybrids. In the case of each of the seven crosses the hybrid-character resembles that of one of the parental forms so closely that the other either escapes observation completely or cannot be detected with certainty. This circumstance is of great importance in the determination and classification of the forms under which the offspring of the hybrids appear. Henceforth in this paper those characters which

are transmitted entire, or almost unchanged in the hybridization, and therefore in themselves constitute the characters of the hybrid, are termed the *dominant*, and those which become latent in the process, *recessive*. The expression “recessive” has been chosen because the characters thereby designated withdraw or entirely disappear in the hybrids, but nevertheless reappear unchanged in their progeny, as will be demonstrated later on.

He then writes about the F_2 generation:

In this generation there reappear, together with the dominant characters, also the recessive ones with their peculiarities fully developed, and this occurs in the definitely expressed average proportion of three to one, so that among each four plants of this generation three display the dominant character and one the recessive. This relates without exception to all the characters which were investigated in the experiments. The angular wrinkled form of the seed, the green colour of the albumen, the white colour of the seed-coats and the flowers, the constrictions of the pods, the yellow colour of the unripe pod, of the stalk, of the calyx, and of the leaf venation, the umbel-like form of the inflorescence, and the dwarfed stem, all reappear in the numerical proportion given, without any essential alteration. *Transitional forms were not observed in any experiment. . . .*

Historical Perspectives

Excerpts from Mendel's Original Paper

F_1 plants are self-fertilized, the result is 315 + 108 round seeds (RR or Rr) and 101 + 32 wrinkled seeds (rr) in the F_2 generation. This is a 423:133 or a 3.18:1.00 phenotypic ratio—very close to the expected 3:1 ratio. So the gene for seed shape is segregating normally. In a similar manner, if we look only at the gene for color, we see that the F_2 ratio of yellow to green seeds is 416:140, or

2.97:1.00—again, very close to a 3:1 ratio. Thus, when two genes are segregating normally according to the rule of segregation, their independent behavior demonstrates the rule of independent assortment (box 2.1).

From the Punnett square in figure 2.15, you can see that because of dominance, all phenotypic classes except the homozygous recessive one—wrinkled, green

Expt. 1. Form of seed.—From 253 hybrids 7,324 seeds were obtained in the second trial year. Among them were 5,474 round or roundish ones and 1,850 angular wrinkled ones. Therefrom the ratio 2.96 to 1 is deduced.

If A be taken as denoting one of the two constant characters, for instance the dominant, a the recessive, and Aa the hybrid form in which both are conjoined, the expression

$$A + 2Aa + a$$

shows the terms in the series for the progeny of the hybrids of two differentiating characters.

Mendel used a notation system different from ours. He designated heterozygotes with both alleles (e.g., Aa) but homozygotes with only one allele or the other (e.g., A for our AA). Thus, whereas he recorded $A + 2Aa + a$, we would record $AA + 2Aa + aa$. Mendel then went on to discuss the dihybrids. He mentions the genotypic ratio of 1:2:1:2:4:2:1:2:1 and the principle of independent assortment:

The fertilized seeds appeared round and yellow like those of the seed parents. The plants raised therefrom yielded seeds of four sorts, which frequently presented themselves in one pod. In all, 556 seeds were yielded by 15 plants, and of those there were:

315 round and yellow,
101 wrinkled and yellow,

108 round and green,
32 wrinkled and green.

Consequently the offspring of the hybrids, if two kinds of differentiating characters are combined therein, are represented by the expression

$$AB + Ab + aB + ab + 2ABb + 2aBb + 2AaB + 4AaBb.$$

(In today's notation, we would write: $AABB + AABb + aaBB + aabb + 2AABb + 2aaBb + 2AaBB + 2Aabb + 4AaBb$.)

This expression is indisputably a combination series in which the two expressions for the characters A and a , B and b are combined. We arrive at the full number of the classes of the series by the combination of the expressions

$$A + 2Aa + a$$

$$B + 2Bb + b$$

(In today's notation we would write

$$AA + 2Aa + aa$$

$$BB + 2Bb + bb.)$$

There is therefore no doubt that for the whole of the characters involved in the experiments the principle applies that *the offspring of the hybrids in which several essentially different characters are combined exhibit the terms of a series of combinations, in which the developmental series for each pair of differentiating characters are united*. It is demonstrated at the same time that *the relation of each pair of different characters in hybrid union is independent of the other differences in the two original parental stocks*.

Table 1 is a summary of all the data Mendel presented on monohybrids (the data from only one dihybrid and one trihybrid cross were presented):

Table 1 Mendel's Data

	Dominant Phenotype	Recessive Phenotype	Ratio
Seed form	5,474	1,850	2.96:1
Cotyledon color	6,022	2,001	3.01:1
Seed coat color	705	224	3.15:1
Pod form	882	299	2.95:1
Pod color	428	152	2.82:1
Flower position	651	207	3.14:1
Stem length	787	277	2.84:1
Total	14,949	5,010	2.98:1

Source: Copyright The Royal Horticultural Society. Taken from the *Journal of the Royal Horticultural Society*, vol. 26. Pg. 1-32. 1901.

seeds—are actually genetically heterogeneous, with phenotypes made up of several genotypes. For example, the dominant phenotypic class, with round, yellow seeds, represents four genotypes: $RRYY$, $RRYy$, $RrYY$, and $RrYy$. When we group all the genotypes by phenotype, we obtain the ratio shown in figure 2.16. Thus, with complete dominance, a self-fertilized dihybrid gives a 9:3:3:1 phe-

notypic ratio in its offspring (F_2). A 1:2:1:2:4:2:1:2:1 genotypic ratio also occurs in the F_2 generation. If the two genes exhibited incomplete dominance or codominance, the latter would also be the phenotypic ratio. What ratio would be obtained if one gene exhibited dominance and the other did not? An example of this case appears in figure 2.17.

BOX 2.2

Overwhelming evidence gathered during this century has proven the correctness of Mendel's conclusions. However, close scrutiny of Mendel's paper has led some to suggest that (1) Mendel failed to report the inheritance of traits that did not show independent assortment and (2) Mendel fabricated numbers. Both these claims are, on the surface, difficult to ignore; both have been countered effectively.

The first claim—that Mendel failed to report crosses involving traits that did not show independent assortment—arises from the observation that all seven traits that Mendel studied do show independent assortment and that the pea plant has precisely seven pairs of chromosomes. For Mendel to have chosen seven genes, one located on each of the seven chromosomes, by chance alone seems extremely unlikely. In fact, the probability would be

$$\frac{7}{7} \times \frac{6}{7} \times \frac{5}{7} \times \frac{4}{7} \times \frac{3}{7} \times \frac{2}{7} \times \frac{1}{7} = 0.006$$

Historical Perspectives

Did Mendel Cheat?

That is, Mendel had less than one chance in one hundred of randomly picking seven traits on the seven different chromosomes. However, L. Douglas and E. Novitski in 1977 analyzed Mendel's data in a different way. To understand their analysis, you have to know that two genes sufficiently far apart on the same chromosome will appear to assort independently (to be discussed in chapter 6). Thus, Mendel's choice of characters showing independent assortment has to be viewed in light of the lengths of the chromosomes. That is, Mendel could have chosen two genes on the same chromosome that would still show independent assortment. In fact, he did. For example, stem length and pod texture (wrinkled or

smooth) are on the fourth chromosome pair in peas. In their analysis, Douglas and Novitski report that the probability of randomly choosing seven characteristics that appear to assort independently is actually between one in four and one in three. So it seems that Mendel did not have to manipulate his choice of characters in order to hide the failure of independent assortment. He had a one in three chance of naively choosing the seven characters that he did, thereby uncovering no deviation from independent assortment.

The second claim—that Mendel fabricated data—comes from a careful analysis of Mendel's paper by R. A. Fisher, a brilliant English statistician and population geneticist. In a paper in 1936, Fisher pointed out two problems in Mendel's work. First, all of Mendel's published data taken together fit their expected ratios better than chance alone would predict. Second, some of Mendel's data fit incorrect expected ratios. This second "error" on Mendel's part came about as follows.

Testcrossing Multihybrids

A simple test of Mendel's rule of independent assortment is the testcrossing of the dihybrid plant. We would predict, for example, that if we crossed an $RrYy$ F_1 individual with an $rryy$ individual, the results would include four phenotypes in a 1:1:1:1 ratio, as shown in figure 2.18. Mendel's data verified this prediction (box 2.2). We will proceed to look at a **trihybrid** cross in order to develop general rules for **multihybrids**.

A trihybrid Punnett square appears in figure 2.19. From this we can see that when a homozygous dominant and a homozygous recessive individual are crossed in the P_1 generation, plants in the F_1 generation are capable of producing eight gamete types. When these F_1 individuals are selfed, they in turn produce F_2 offspring of twenty-seven different genotypes in a ratio of sixty-fourths. By extrapolating from the monohybrid through the trihybrid, or simply by the rules of probability, we can construct table 2.3, which contains the rules for F_1 gamete

production and F_2 zygote formation in a multihybrid cross. For example, from this table we can figure out the F_2 offspring when a dodecahybrid (twelve segregating genes: $AA BB CC \dots LL \times aa bb cc \dots ll$) is selfed. The F_1 organisms in that cross will produce gametes with 2^{12} , or 4,096, different genotypes. The proportion of homozygous recessive offspring in the F_2 generation is $1/(2^n)^2$ where $n = 12$, or 1 in 16,777,216. With complete dominance, there will be 4,096 different phenotypes in the F_2 generation. If dominance is incomplete, there can be 3^{12} , or 531,441, different phenotypes in the F_2 generation.

GENOTYPIC INTERACTIONS

Often, several genes contribute to the same phenotype. An example occurs in the combs of fowl (fig. 2.20). If we cross a rose-combed hen with a pea-combed rooster (or vice versa), all the F_1 offspring are walnut-combed. If we

Mendel determined whether a dominant phenotype in the F_2 generation was a homozygote or a heterozygote by self-fertilizing it and examining ten offspring. In an F_2 generation composed of $1AA:2Aa:1aa$, he expected a 2:1 ratio of heterozygotes to homozygotes within the dominant phenotypic class. In fact, this ratio is not precisely correct because of the problem of misclassification of heterozygotes. It is probable that some heterozygotes will be classified as homozygotes because all their offspring will be of the dominant phenotype. The probability that one offspring from a selfed Aa individual has the dominant phenotype is $3/4$, or 0.75: the probability that ten offspring will be of the dominant phenotype is $(0.75)^{10}$ or 0.056. Thus, Mendel misclassified heterozygotes as dominant homozygotes 5.6% of the time. He should have expected a 1.89:1.11 ratio instead of a 2:1 ratio to demonstrate segregation. Mendel classified 600 plants this way in one cross and got a ratio of 201 homozygous to 399 heterozygous offspring.

This is an almost perfect fit to the presumed 2:1 ratio and thus a poorer fit to the real 1.89:1.11 ratio. This bias is consistent and repeated in Mendel's trihybrid analysis.

Fisher, believing in Mendel's basic honesty, suggested that Mendel's data do not represent an experiment but more of a hypothetical demonstration. In 1971, F. Weiling published a more convincing case in Mendel's defense. Pointing out that the data of Mendel's rediscoverers are also suspect for the same reason, he suggested that the problem lies with the process of pollen formation in plants, not with the experimenters. In an Aa heterozygote, two A and two a cells develop from a pollen mother cell. These cells tend to stay together on the anther. Thus, pollen cells do not fertilize in a strictly random fashion. A bee is more likely to take equal numbers of A and a pollen than chance alone would predict. The result is that the statistics Fisher used are not applicable. By using a different statistic, Weiling showed that, in fact, Mendel need not have manipu-

lated any numbers (nor would have his rediscoverers) in order to get data that fit the expected ratios well. By the same reasoning, very little misclassification of heterozygotes would have occurred.

More recently, Weiling and others have made several additional points. First, for Mendel to be sure of ten offspring, he probably examined more than ten, and thus he probably kept his misclassification rate lower than 5.6%. Second, despite Fisher's brilliance as a statistician, several have made compelling arguments that Fisher's statistical analyses were incorrect. In other words, for subtle statistical reasons, many of his analyses involved methods and conclusions that were in error.

We conclude that there is no compelling evidence to suggest that Mendel in any way manipulated his data to demonstrate his rules. In fact, taking into account what is known about him personally, it is much more logical to believe that he did not "cheat."

cross the hens and roosters of this heterozygous F_1 group, we will get, in the F_2 generation, walnut-, rose-, pea-, and single-combed fowl in a ratio of 9:3:3:1. Can you figure out the genotypes of this F_2 population before reading further? An immediate indication that two allelic pairs are involved is the fact that the 9:3:3:1 ratio appeared in the F_2 generation. As we have seen, this ratio comes about

when we cross dihybrids in which both genes have alleles that control traits with complete dominance.

Figure 2.21 shows the analysis of this cross. When dominant alleles of both genes are present in an individual ($R-P-$), the walnut comb appears. (The dash indicates any second allele; thus, $R-P-$ could be $RRPP$, $RrPP$, $RRPp$, or $RrPp$.) A dominant allele of the rose gene ($R-$) with

Table 2.3 Multihybrid Self-Fertilization, Where n Equals Number of Genes Segregating Two Alleles Each

	Monohybrid $n = 1$	Dihybrid $n = 2$	Trihybrid $n = 3$	General Rule
Number of F_1 gametic genotypes	2	4	8	2^n
Proportion of recessive homozygotes among the F_2 individuals	$1/4$	$1/16$	$1/64$	$1/(2^n)^2$
Number of different F_2 phenotypes, given complete dominance	2	4	8	2^n
Number of different genotypes (or phenotypes, if no dominance exists)	3	9	27	3^n

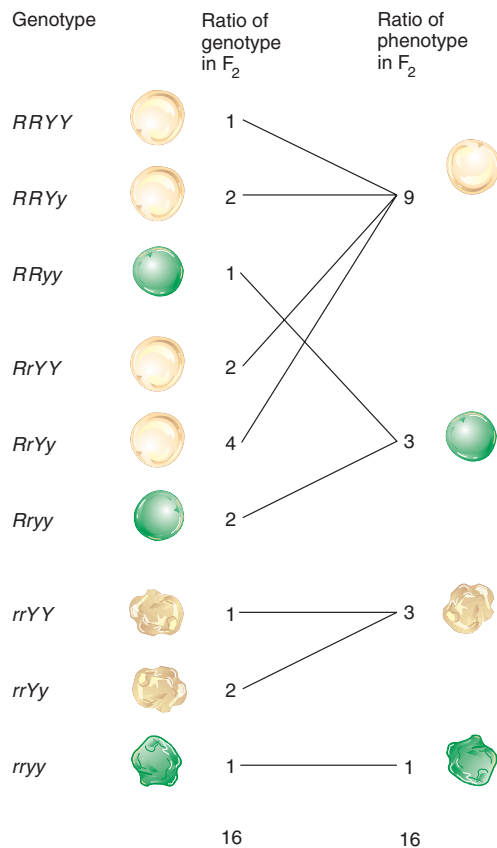


Figure 2.16 The phenotypic and genotypic ratios of the offspring of dihybrid peas.

recessive alleles of the pea gene (*pp*) gives a rose comb. A dominant allele of the pea gene (*P*-) with recessive alleles of the rose gene (*rr*) gives pea-combed fowl. When both genes are homozygous for the recessive alleles, the fowl are single-combed. Thus, a 9:3:3:1 F₂ ratio arises from crossing dihybrid individuals even though different expressions of the same phenotypic characteristic, the comb, are involved. In our previous 9:3:3:1 example (see fig. 2.15), we dealt with two separate characteristics: shape and color of peas.

In corn (or maize, *Zea mays*), several different field varieties produce white kernels on the ears. In certain crosses, two white varieties will result in an F₁ generation with all purple kernels. If plants grown from these purple kernels are selfed, the F₂ individuals have both purple and white kernels in a ratio of 9:7. How can we explain this? We must be dealing with the offspring of dihybrids with each gene segregating two alleles, because the ratio is in sixteenths. Furthermore, we can see that the F₂ 9:7 ratio is a variation of the 9:3:3:1 ratio. The 3, 3, and 1 categories here are producing the same phenotype and thus make up 7/16 of the F₂ offspring. Figure 2.22 outlines the cross. We can see from this figure that the purple color appears only when dominant alleles of both genes are present. When one or both genes have only recessive alleles, the kernels will be white.

Epistasis

The color of corn kernels illustrates the concept of **epistasis**, the interaction of nonallelic genes in the formation

Figure 2.17 Independent assortment of two blood systems in human beings. In the ABO system, the *I^A* and *I^B* alleles are codominant. In a simplified view of the Rhesus system, the Rh⁺ phenotype (*D* allele) is dominant to the Rh⁻ phenotype (*d* allele).

P₁		<i>I^AI^ADD</i>		X	<i>I^BI^Bdd</i>		
F₁		<i>I^AI^BDd</i>		(F ₁ X F ₁)			
		Male					
		<i>I^AD</i>	<i>I^Ad</i>	<i>I^BD</i>	<i>I^Bd</i>		
F₂	<i>I^AD</i>	<i>I^AI^ADD</i>	<i>I^AI^ADd</i>	<i>I^AI^BDD</i>	<i>I^AI^BDd</i>		
	<i>I^Ad</i>	<i>I^AI^ADd</i>	<i>I^AI^Add</i>	<i>I^AI^BDd</i>	<i>I^AI^Bdd</i>		
	<i>I^BD</i>	<i>I^AI^BDD</i>	<i>I^AI^BDd</i>	<i>I^BI^BDD</i>	<i>I^BI^BDd</i>		
	<i>I^Bd</i>	<i>I^AI^BDd</i>	<i>I^AI^Bdd</i>	<i>I^BI^BDd</i>	<i>I^BI^Bdd</i>		
F₂	Phenotype	A Rh ⁺	A Rh ⁻	B Rh ⁺	B Rh ⁻	AB Rh ⁺	AB Rh ⁻
Summary	Frequency	3	1	3	1	6	2

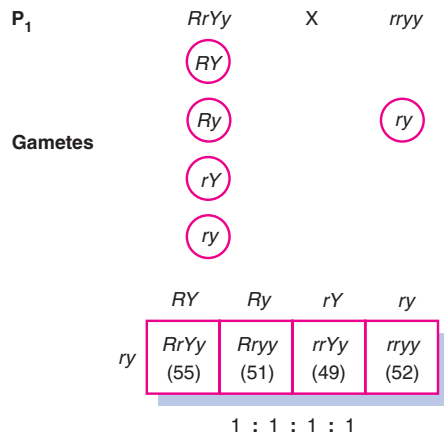


Figure 2.18 Testcross of a dihybrid. A 1:1:1:1 ratio is expected in the offspring.

of the phenotype. This is a process analogous to dominance among alleles of one gene. For example, the recessive apterous (wingless) gene in fruit flies is epistatic to any gene that controls wing characteristics; hairy wing is hypostatic to apterous (that is, the recessive apterous gene, when homozygous, masks the presence of the hairy wing gene, because, obviously, without wings, no

wing characteristics can be expressed). Note that the genetic control of comb type in fowl does not involve epistasis. There are no allelic combinations at one locus that mask genotypes at another locus: the 9:3:3:1 ratio is not an indication of epistasis. To illustrate further the principle of epistasis, we can look at the control of coat color in mice.

In one particular example, a pure-breeding black mouse is crossed with a pure-breeding albino mouse (pure white because all pigment is lacking); all of the offspring are agouti (the typical brownish-gray mouse color). When the F_1 agouti mice are crossed with each other, agouti, black, and albino offspring appear in the F_2 generation in a ratio of 9:3:4. What are the genotypes in this cross? The answer appears in figure 2.23. By now it should be apparent that the F_2 ratio of 9:3:4 is also a variant of the 9:3:3:1 ratio; it indicates epistasis in a dihybrid cross. What is the mechanism producing this 9:3:4 ratio? Of a potential 9:3:3:1 ratio, one of the 3/16 classes and the 1/16 class are combined to create a 4/16 class. Any genotype that includes $c^a c^a$ will be albino, masking the A gene, but as long as at least one dominant C allele is present, the A gene can express itself. Mice with dominant alleles of both genes ($A-C-$) will have the agouti color, whereas mice that are homozygous recessive at the A gene ($aaC-$) will be black. So, at the A gene, A for agouti

P_1	$AA BB CC \times aa bb cc$							
F_1	$Aa Bb Cc \times \text{Self}$							
	ABC	ABc	AbC	Abc	aBC	aBc	abC	abc
ABC	$AA BB CC$	$AA BB Cc$	$AA Bb CC$	$AA Bb Cc$	$Aa BB CC$	$Aa BB Cc$	$Aa Bb CC$	$Aa Bb Cc$
ABc	$AA BB Cc$	$AA BB cc$	$AA Bb Cc$	$AA Bb cc$	$Aa BB Cc$	$Aa BB cc$	$Aa Bb Cc$	$Aa Bb cc$
AbC	$AA Bb CC$	$AA Bb Cc$	$AA bb CC$	$AA bb Cc$	$Aa Bb CC$	$Aa Bb Cc$	$Aa bb CC$	$Aa bb Cc$
Abc	$AA Bb Cc$	$AA Bb cc$	$AA bb Cc$	$AA bb cc$	$Aa Bb Cc$	$Aa Bb cc$	$Aa bb Cc$	$Aa bb cc$
aBC	$Aa BB CC$	$Aa BB Cc$	$Aa Bb CC$	$Aa Bb Cc$	$aa BB CC$	$aa BB Cc$	$aa Bb CC$	$aa Bb Cc$
aBc	$Aa BB Cc$	$Aa BB cc$	$Aa Bb Cc$	$Aa Bb cc$	$aa BB Cc$	$aa BB cc$	$aa Bb Cc$	$aa Bb cc$
abC	$Aa Bb CC$	$Aa Bb Cc$	$Aa bb CC$	$Aa bb Cc$	$aa Bb CC$	$aa Bb Cc$	$aa bb CC$	$aa bb Cc$
abc	$Aa Bb Cc$	$Aa Bb cc$	$Aa bb Cc$	$Aa bb cc$	$aa Bb Cc$	$aa Bb cc$	$aa bb Cc$	$aa bb cc$

Figure 2.19 Trihybrid cross.

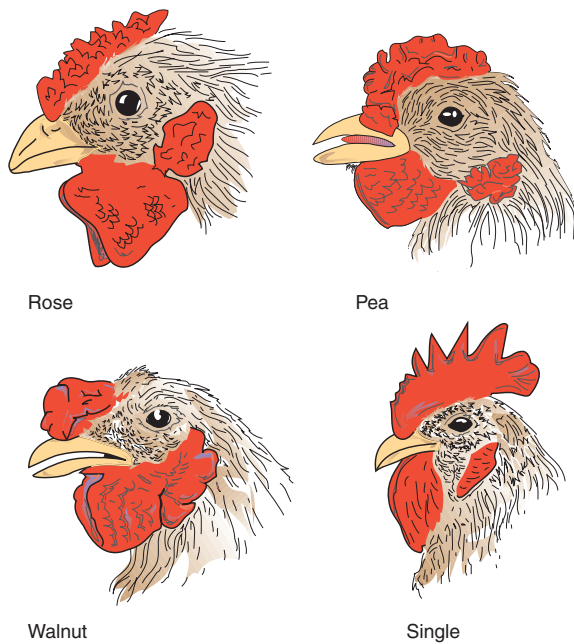


Figure 2.20 Four types of combs in fowl.

P₁

Rose comb $RRpp$ × Pea comb $rrPP$

F₁

Walnut comb $RrPp$
 $F_1 \times F_1$

F₂	<i>RP</i>	<i>Rp</i>	<i>rP</i>	<i>rp</i>
<i>RP</i>	$RRPP$ Walnut	$RRPp$ Walnut	$RrPP$ Walnut	$RrPp$ Walnut
<i>Rp</i>	$RRPp$ Walnut	$RRpp$ Rose	$RrPp$ Walnut	$Rrpp$ Rose
<i>rP</i>	$RrPP$ Walnut	$RrPp$ Walnut	$rrPP$ Pea	$rrPp$ Pea
<i>rp</i>	$RrPp$ Walnut	$Rrpp$ Rose	$rrPp$ Pea	$rrpp$ Single

F₂ Summary

Walnut : Rose : Pea : Single
9 : 3 : 3 : 1

Figure 2.21 Independent assortment in the determination of comb type in fowl.

P₁ First white variety $AAbb$ × Second white variety $aaBB$

F₁ Purple
 $AaBb \times \text{Self}$

F₂	<i>AB</i>	<i>Ab</i>	<i>aB</i>	<i>ab</i>
<i>AB</i>	$AABB$	$AABb$	$AaBB$	$AaBb$
<i>Ab</i>	$AABb$	$AAbb$	$AaBb$	$Aabb$
<i>aB</i>	$AaBB$	$AaBb$	$aaBB$	$aaBb$
<i>ab</i>	$AaBb$	$Aabb$	$aaBb$	$aabb$

F₂ Summary

Purple : White
9 : 7

Figure 2.22 Color production in corn.

is dominant to a for black. The albino gene (c^a), when homozygous, is epistatic to the A gene; the A gene is hypostatic to the gene for albinism.

Mechanism of Epistasis

In this case, the physiological mechanism of epistasis is known. The pigment melanin is present in both the black and agouti phenotypes. The agouti is a modified black hair in which yellow stripes (the pigment phaeomelanin) have been added. Thus, with melanin present, agouti is dominant. Without melanin, we get an albino regardless of the genotype of the agouti gene because both agouti and black depend on melanin. Albinism is the result of one of several defects in the enzymatic pathway for the synthesis of melanin (fig. 2.24).

Knowing that epistatic modifications of the 9:3:3:1 ratio come about through gene interactions at the biochemical level, we can look for a biochemical explanation for the 9:7 ratio in corn kernel color (fig. 2.22). Two possible mechanisms for a 9:7 ratio are shown in figure 2.25. Either a two-step process takes a precursor molecule and turns it into purple pigment, or two precursors that must be converted to final products then combine to produce purple pigment. The dominant alleles from the two genes control the two steps in the process. Recessive alleles are ineffective. Thus, dominants are necessary for both steps to complete the pathways for a purple pig-

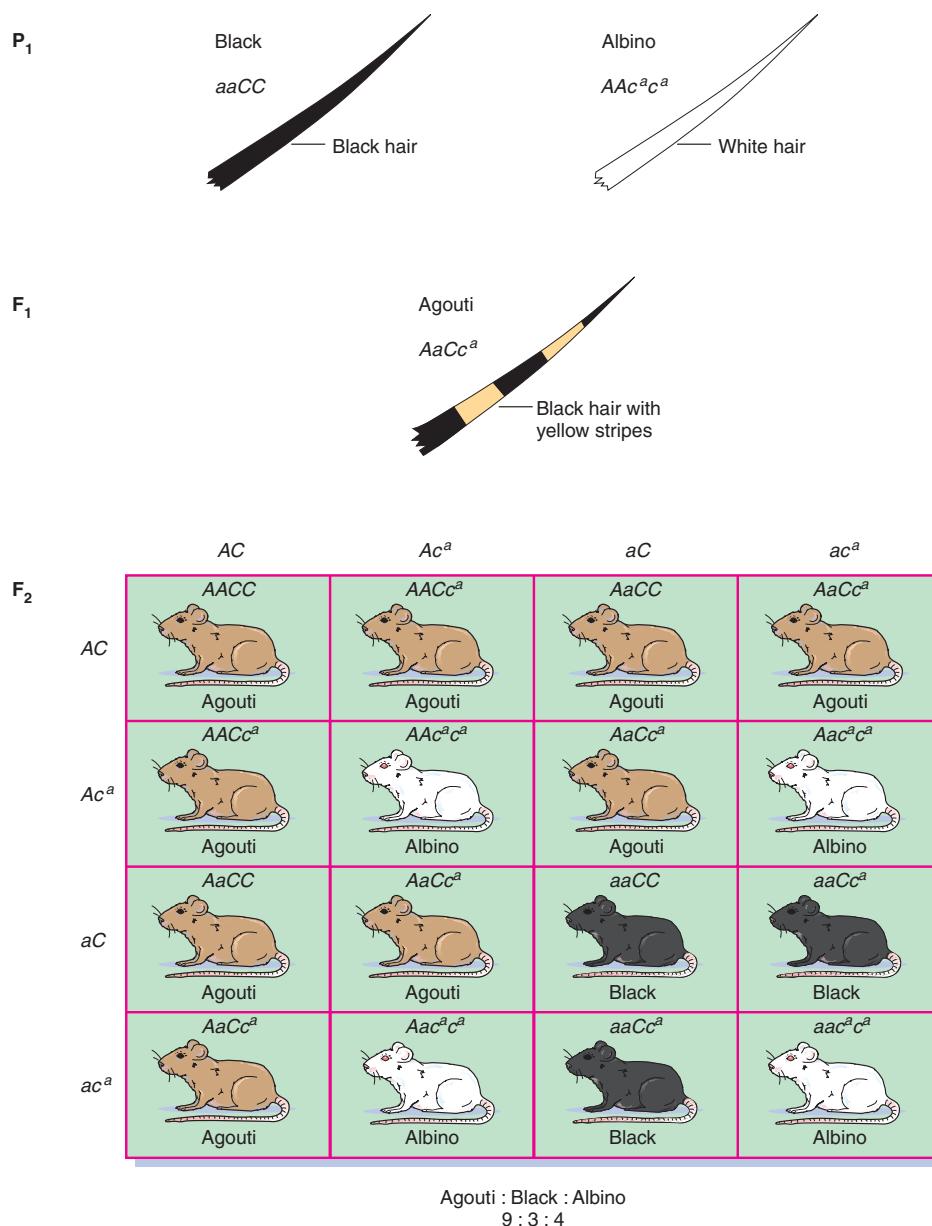


Figure 2.23 Epistasis in the coat color of mice.

ment. Stopping the process at any point prevents the production of purple color.

Another example of epistasis occurs in the snapdragon (*Antirrhinum majus*). There, a gene called *nivea* has alleles that determine whether any pigment is produced; the *nn* genotype prevents pigment production, whereas the *NN* or *Nn* genotypes permit pigment color genes to express themselves. The *eosinea* gene controls the production of a red anthocyanin pigment. In the

presence of the *N* allele of the *nivea* gene, the genotypes *EE* or *Ee* of the *eosinea* gene produce red flowers; the *ee* genotype produces pink flowers. When dihybrids are self-fertilized, red-, pink-, and white-flowered plants are produced in a ratio of 9:3:4 (fig. 2.26). The epistatic interaction is the *nn* genotype masking the expression of alleles at the *eosinea* gene. In other words, regardless of the genotypes of the *eosinea* gene (*EE*, *Ee*, or *ee*), the flowers will be white if the *nivea* gene has the *nn*

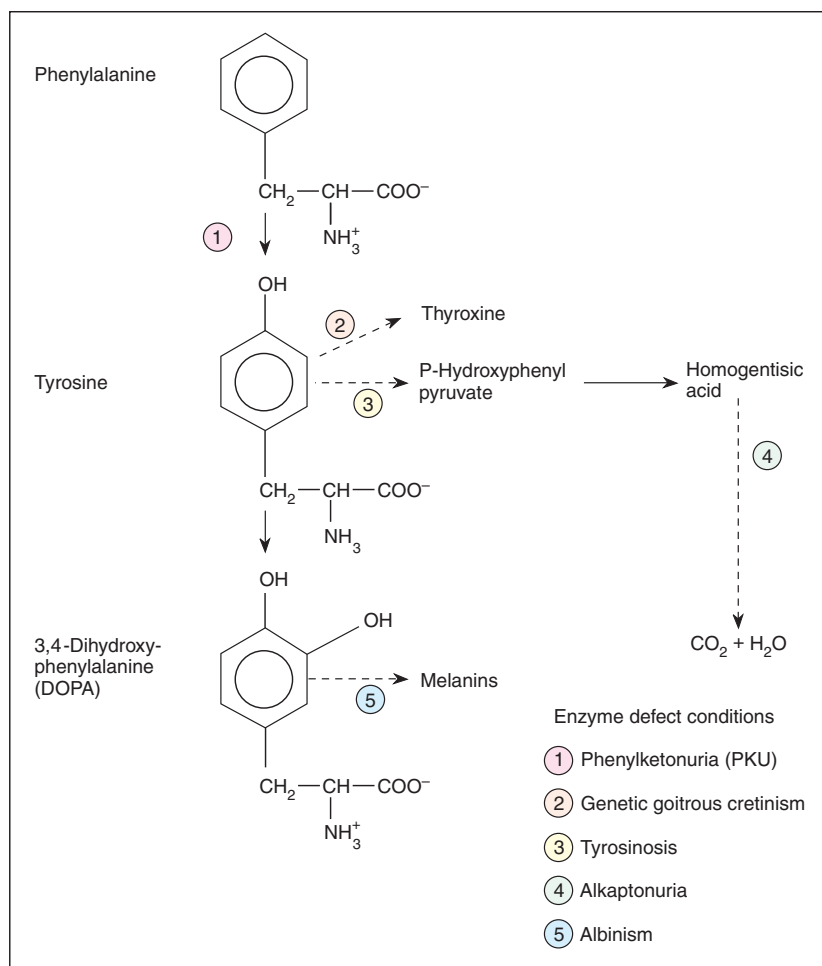


Figure 2.24 In humans, errors in melanin synthesis produce different physical conditions and diseases, depending on which part of the tyrosine (an amino acid) metabolic pathway is disrupted. The broken arrows indicate that there is more than one step in the pathways; the conditions listed occur only in homozygous recessives.

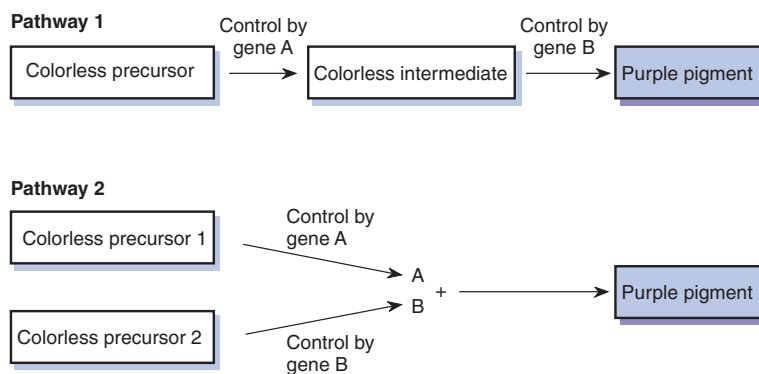


Figure 2.25 Possible metabolic pathways of color production that would yield 9:7 ratios in the F_2 generation of a self-fertilized dihybrid.

combination of alleles. Thus, *nivea* is epistatic to *eosinea*, and *eosinea* is hypostatic to *nivea*. (We should add that at least seven major colors occur in snapdragons, along with subtle shade differences, all genetically controlled by the interactions of at least seven genes.)

Other types of epistatic interactions occur in other organisms. Table 2.4 lists several. We do not know the exact physiological mechanisms in many cases, especially when developmental processes are involved (e.g., size and shape). However, from an analysis of crosses, we can know the number of genes involved and the general nature of their interactions.

BIOCHEMICAL GENETICS

Inborn Errors of Metabolism

The examples of mouse coat color, corn kernel color, and snapdragon flower petal color demonstrate that genes control the formation of enzymes, proteins that control the steps in biochemical pathways. For the most part, dominant alleles control functioning enzymes that catalyze biochemical steps. Recessive alleles often produce nonfunctioning enzymes that cannot catalyze specific steps. Often a heterozygote is normal because one allele produces a functional enzyme; usually only half the enzyme quantity of the dominant homozygote is enough. The study of the relationship between genes and enzymes is generally called **biochemical genetics** because it involves the genetic control of biochemical pathways. A. E. Garrod, a British physician, pointed out this general concept of human gene action in *Inborn Errors of Metabolism*, published in 1909. Only nine years after Mendel was rediscovered, Garrod described several human conditions, such as albinism and alkaptonuria, that occur in individuals who are homozygous for recessive alleles (see fig. 2.24).

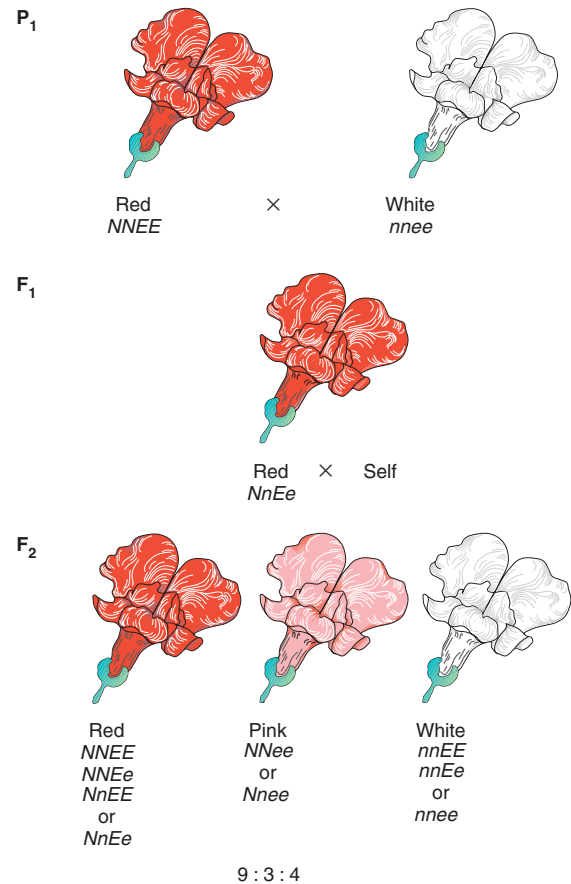


Figure 2.26 Flower color inheritance in snapdragons. This is an example of epistasis: an *nn* genotype masks the expression of alleles (*EE*, *Ee*, or *ee*) at the *eosinea* gene.

Table 2.4 Some Examples of Epistatic Interactions Among Alleles of Two Genes

Characteristic	Phenotype of F ₁ Dihybrid (<i>AaBb</i>)	Phenotypic F ₂ Ratio
Corn and sweet pea color	Purple	Purple:white 9:7
Mouse coat color	Agouti	Agouti:black:albino 9:3:4
Shepherd's purse seed capsule shape	Triangular	Triangular:oval 15:1
Summer squash shape	Disk	Disk:sphere:elongate 9:6:1
Fowl color	White	White:colored 13:3

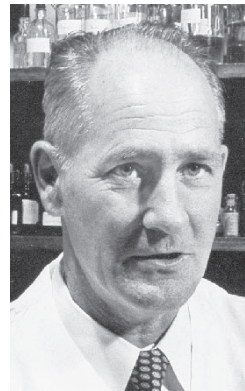
For example, people normally degrade homogentisic acid (alkapton) into maleylacetoacetic acid. Persons with the disease alkaptonuria are homozygous for a nonfunctional form of the enzyme essential to the process: homogentisic acid oxidase, found in the liver. Absence of this enzyme blocks the degradation reaction so that homogentisic acid builds up. This acid darkens upon oxidation. Thus, affected persons can be identified by the black color of their urine after its exposure to air. Eventually, alkaptonuria causes problems in the joints and a darkening of cartilage that is visible in the ears and the eye sclera.

One-Gene-One-Enzyme Hypothesis

Pioneering work in the concept that genes control the production of enzymes, which in turn control the steps in biochemical pathways, was done by George Beadle and Edward Tatum, who eventually shared the Nobel Prize for their work. They not only put forth the **one-gene-one-enzyme hypothesis**, but also used mutants to work out the details of biochemical pathways. In 1941, Beadle and Tatum were the first scientists to isolate mutants with nutritional requirements that defined steps in a biochemical pathway. In the early 1940s, they united the fields of biochemistry and genetics by using strains of a bread mold with specific nutritional requirements to discover the steps in biochemical pathways in that organism.

Through this century, the study of mutations has been the driving force in genetics. The process of mutation produces alleles that differ from the wild-type and shows us that a particular aspect of the phenotype is under genetic control. Beadle and Tatum used mutants to work out the steps in the biosynthesis of niacin (vitamin B₃) in pink bread mold, *Neurospora crassa*.

Normally, *Neurospora* synthesizes niacin via the pathway shown in figure 2.27. Beadle and Tatum isolated mutants that could not grow unless niacin was provided in



George W. Beadle
(1903–89). Courtesy of the
Archives, California Institute of
Technology.



Edward L. Tatum
(1909–75). Courtesy of the
Proceedings for the National
Academy of Sciences.

the culture medium; these mutants had enzyme deficiencies in the synthesizing pathway that ends with niacin. Thus, although wild-type *Neurospora* could grow on a medium without additives, the mutants could not. Beadle and Tatum had a general idea, based on the structure of niacin, as to what substances would be in the niacin biosynthesis pathway. They could thus make educated guesses as to what substances they might add to the culture medium to enable the mutants to grow. Mutant B (table 2.5), for example, could grow if given niacin or, alternatively, 3-hydroxyanthranilic acid. It could not grow if given only kynurenine. Thus, Beadle and Tatum knew that the B mutation affected the pathway between kynurenine and 3-hydroxyanthranilic acid. Similarly, mutant A could grow if given 3-hydroxyanthranilic acid or kynurenine instead of niacin. Therefore, these two prod-

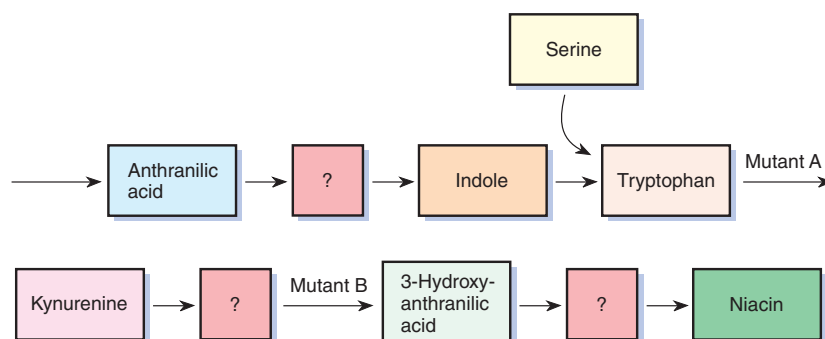


Figure 2.27 Pathway of niacin synthesis in *Neurospora*. Each arrow represents an enzyme-mediated step. Each question mark represents a presumed but (at the time Beadle and Tatum were working) unknown compound.

Table 2.5 Growth Performance of *Neurospora* Mutants (plus sign indicates growth; minus sign indicates no growth)

	Additive			
	Tryptophan	Kynurenine	3-Hydroxyanthranilic Acid	Niacin
Wild-type	+	+	+	+
Mutant A	—	+	+	+
Mutant B	—	—	+	+

ucts must be in the pathway *after* the step interrupted in mutant A. Conversely, since neither of these mutant organisms could grow when given only tryptophan, Beadle and Tatum knew that tryptophan occurred in the pathway before the steps with the deficient enzymes. By this type of analysis, they discovered the steps in several biochemical pathways of *Neurospora*. Many biochemical pathways are similar in a huge range of organisms, and thus Beadle and Tatum's work was of general importance. (We will spend more time studying *Neurospora* in chapter 6.)

Beadle and Tatum could further verify their work by observing which substances accumulated in the mutant organisms. If a biochemical pathway is blocked at a certain point, then the substrate at that point cannot convert into the next product, and it builds up in the cell. For example, in the niacin pathway (fig. 2.27), if a block occurs just after 3-hydroxyanthranilic acid, that substance will build up in the cell because it cannot convert into the next substance on the way to niacin.

This analysis could be misleading, however, if the built-up substance is being "siphoned off" into other biochemical pathways in the cell. Also, the cell might attempt to break down or sequester toxic substances. This would mean there might not be an obvious buildup of the substance just before the blocked step.

Beadle and Tatum concluded from their studies that one gene controls the production of one enzyme. The one-gene-one-enzyme hypothesis is an oversimplification that we will clarify later in the book. As a rule of thumb, however, the hypothesis is valid, and it has served to direct attention to the functional relationship between genes and enzymes in biochemical pathways.

Although a change in a single enzyme usually disrupts a single biochemical pathway, it frequently has more than one effect on phenotype. Multiple effects are referred to as **pleiotropy**. A well-known example is sickle-cell anemia, caused by a mutation in the gene for the β chain of the hemoglobin molecule. In a homozygote, this mutation causes a sickling of red blood cells (fig. 2.28). The sickling of these cells has two major ramifications.

First, the liver destroys the sickled cells, causing anemia. The phenotypic effects of this anemia include physical weakness, slow development, and hypertrophy of the bone marrow, resulting in the "tower skull" seen in some of those afflicted with the disease. The second major effect of sickle-cell anemia is that the sickled cells interfere with capillary blood flow, clumping together and resulting in damage to every major organ. The individual can suffer pain, heart failure, rheumatism, and other ill effects. Hence, a single mutation shows itself in many aspects of the phenotype.



Figure 2.28 Sickle-shaped red blood cells from a person with sickle-cell anemia. Red blood cells are about 7 to 8 μm in diameter. (Courtesy of Dr. Patricia N. Farnsworth.)

S U M M A R Y

STUDY OBJECTIVE 1: To understand that genes are discrete units that control the appearance of an organism 17–18

Genes control phenotypic traits such as size and color. They are inherited as discrete units.

STUDY OBJECTIVE 2: To understand Mendel's rules of inheritance: segregation and independent assortment 18–22

Higher organisms contain two alleles of each gene, but only one allele enters each gamete. Zygote formation restores the double number of alleles in the cell. This is Mendel's rule of segregation.

Alleles of different genes segregate independently of each other. Mendel was the first to recognize the 3:1 phenotypic ratio as a pattern of inheritance; the 9:3:3:1 ratio demonstrates independent assortment in hybrids. Mendel was successful in his endeavor because he performed careful experiments using discrete characteristics, large numbers of offspring, and an organism (the pea plant) amenable to controlled fertilizations.

STUDY OBJECTIVE 3: To understand that dominance is a function of the interaction of alleles; similarly, epistasis is a function of the interaction of nonallelic genes 22–37

There can be many alleles for one gene, although each individual organism has only two alleles for each gene. A phenotype is dominant if it is expressed when one or two copies of its allele are present (heterozygote or homozygote). Dominance depends, however, on the level of the phenotype one looks at.

Genes usually control the production of enzymes, which control steps in metabolic pathways. Many human metabolic diseases are due to homozygosity of an allele that produces a nonfunctioning enzyme.

Nonallelic genes can interact in producing a phenotype so that alleles of one gene mask the expression of alleles of another gene. This process, termed *epistasis*, alters the expected phenotypic ratios.

STUDY OBJECTIVE 4: To define how genes generally control the production of enzymes and thus the fate of biochemical pathways 37–39

Beadle and Tatum used mutants with mutations in the niacin biosynthesis pathway to work out the steps in the pathway. A single mutation can have many phenotypic effects (pleiotropy).

S O L V E D P R O B L E M S

PROBLEM 1: In corn, rough sheath (*rs*) is recessive to smooth sheath (*Rs*), midrib absent (*mrl*) is recessive to midrib present (*Mrl*), and crinkled leaf (*cr*) is recessive to smooth leaf (*Cr*). (Alleles are named for the mutants, which are all recessive.) What are the results of testcrossing a trihybrid?

Answer: The trihybrid has the genotype *RsrS Mrlmrl Crcr*. This parent is capable of producing eight different gamete types in equal frequencies, all combinations of one allele from each gene (*Rs Mrl Cr*; *Rs Mrl cr*; *Rs mrl Cr*; *Rs mrl cr*; *rs Mrl Cr*; *rs Mrl cr*; *rs mrl Cr*; and *rs mrl cr*). In a testcross, the other parent is a recessive homozygote with the genotype *rsrs mrlmrl crcr*; capable of producing only one type of gamete, with the alleles *rs mrl cr*. Thus, this cross can produce zygotes of eight different genotypes (and phenotypes), one for each of the gamete types of the trihybrid parent: *RsrS Mrlmrl Crcr* (smooth sheath, midrib present, smooth leaf); *RsrS Mrlmrl crcr* (smooth sheath, midrib present, crinkled leaf); *RsrS mrlmrl Crcr* (smooth sheath, midrib absent, smooth leaf); *RsrS mrlmrl*

crcr (smooth sheath, midrib absent, crinkled leaf); *rsrs Mrlmrl Crcr* (rough sheath, midrib present, smooth leaf); *rsrs Mrlmrl crcr* (rough sheath, midrib present, crinkled leaf); *rsrs mrlmrl Crcr* (rough sheath, midrib absent, smooth leaf); and *rsrs mrlmrl crcr* (rough sheath, midrib absent, crinkled leaf). Each should make up one-eighth of the total number of offspring.

PROBLEM 2: Summer squash come in three shapes: disk, spherical, and elongate. In one experiment, researchers crossed two squash plants with disk-shaped fruits. The first 160 seeds planted from this cross produced plants with fruit shapes as follows: 89 disk, 61 sphere, and 10 elongate. What is the mode of inheritance of fruit shape in summer squash?

Answer: The numbers are very close to a ratio of 90:60:10, or 9:6:1, an epistatic variant of the 9:3:3:1, with the two 3/16ths categories combined. If this is the case, then the parent plants with disk-shaped fruits were dihybrids (*AaBb*). Among the offspring, 9/16ths had disk-

shaped fruit, indicating that it takes at least one dominant allele of each gene to produce disk-shaped fruits ($A-B-$: $AABB$, $AaBB$, $AABb$, or $AaBb$). The 1/16th category of plants with elongate fruits indicates that this fruit shape occurs in homozygous recessive plants ($aabb$). The plants with spherical fruit are thus plants with a dominant allele of one gene but a homozygous recessive combination at the other gene ($AAbb$, $Aabb$, $aaBB$, or $aaBb$). In summary, then, two genes combine to control fruit shape in summer squash. The epistatic interactions between the two genes produce a 9:6:1 ratio of offspring phenotypes when dihybrids are crossed.

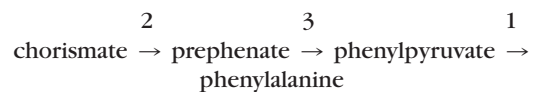
PROBLEM 3: A geneticist studying the pathway of synthesis of phenylalanine in *Neurospora* isolated several mutants that require phenylalanine to grow. She tested whether

	Additive			
	Phenylpyruvate	Prephenate	Chorismate	Phenylalanine
Wild-type	+	+	+	+
Mutant 1	—	—	—	+
Mutant 2	+	+	—	+
Mutant 3	+	—	—	+

each mutant would grow when provided additives that she believed were in the pathway of phenylalanine synthesis (see table); a plus indicates growth and minus indicates the lack of growth in the three mutants tested.

Where in the pathway to phenylalanine synthesis does each of the additives belong, if at all?

Answer: The wild-type grows in the presence of all additives. This is not surprising since the wild-type can grow, by definition, in the absence of all the additives because it can synthesize phenylalanine *de novo*. Mutant 1 cannot grow in the presence of any additive except phenylalanine, indicating that its mutation affects the step just before the end of the pathway at phenylalanine. In other words, each of the other additives occurs in the phenylalanine pathway before the point of the mutation in mutant 1. Mutant 2 can grow if given any additive but chorismate, indicating that chorismate is at the beginning of the pathway, and the mutation affects the pathway just after that step. Finally, mutant 3 can grow if given phenylpyruvate or phenylalanine, indicating that its mutation affects the step before phenylpyruvate and phenylalanine, but after the earlier part of the pathway. Putting all of this information together indicates that the pathway to phenylalanine, with mutants indicated, is:



EXERCISES AND PROBLEMS *

SEGREGATION

1. Mendel crossed tall pea plants with dwarf ones. The F_1 plants were all tall. When these F_1 plants were selfed to produce the F_2 generation, he got a 3:1 tall-to-dwarf ratio in the offspring. Predict the genotypes and phenotypes and relative proportions of the F_3 generation produced when the F_2 generation was selfed.
2. What properties of fruit flies and corn made them the organisms of choice for geneticists during most of the first half of the twentieth century? (Molecular geneticists have made great strides working with bacteria and viruses. You could begin thinking at this point about the properties that have made these organisms so valuable to geneticists.)
3. State precisely the rules of segregation and independent assortment. (See also the Exercises and Problems section on Independent Assortment.)
4. In *Drosophila*, a cross between a dark-bodied fly and a tan-bodied fly yields seventy-six tan and eighty dark flies. Diagram the cross.

5. If two black mice are crossed, ten black and three white mice result.
 - a. Which allele is dominant?
 - b. Which allele is recessive?
 - c. What are the genotypes of the parents?
6. In *Drosophila*, two red-eyed flies mate and yield 110 red-eyed and 35 brown-eyed offspring. Diagram the cross and determine which allele is dominant.

DOMINANCE IS NOT UNIVERSAL

7. Explain how Tay-Sachs disease can be both a recessive and an incomplete dominant trait. What are the differences between incomplete dominance and codominance?
8. How does the biochemical pathway in figure 2.13 explain how alleles I^A and I^B are codominant, yet both dominant to allele i ?

*Answers to selected exercises and problems are on page A-1.

9. Two short-eared pigs are mated. In the progeny, three have no ears, seven have short ears, and four have long ears. Explain these results by diagramming the cross.
10. A plant with red flowers is crossed with a plant with white flowers. All the progeny are pink. When the plants with pink flowers are crossed, the progeny are eleven red, twenty-three pink, and twelve white. What is the mode of inheritance of flower color?

NOMENCLATURE

11. In fruit flies, a new dominant trait, washed eye, was discovered. Describe different ways of naming the alleles of the washed-eye gene.
12. The following is a list of ten genes in fruit flies, each with one of its alleles given. Are the alleles shown dominant or recessive? Are they mutant or wild-type? What is the alternative allele for each? Is the alternative allele dominant or recessive in each case?

Name of Gene	Allele
<i>yellow</i>	y^+
<i>Hairy wing</i>	<i>Hw</i>
<i>Abruptex</i>	Ax^+
<i>Confluens</i>	<i>Co</i>
<i>raven</i>	rv^+
<i>downy</i>	<i>dow</i>
<i>Minute(2)e</i>	$M(2)e^+$
<i>Jammed</i>	<i>J</i>
<i>tufted</i>	tuf^+
<i>burgundy</i>	<i>bur</i>

MULTIPLE ALLELES

13. In the ABO blood system in human beings, alleles I^A and I^B are codominant, and both are dominant to the i allele. In a paternity dispute, a type AB woman claimed that one of four men, each with different blood types, was the father of her type A child. Which of the following could be the blood type of the father of the child on the basis of the evidence given?
- Type A
 - Type B
 - Type O
 - Type AB
14. Under what circumstances can the phenotypes of the ABO system be used to refute paternity?
15. In blood transfusions, one blood type is called the "universal donor" and one the "universal recipient" because of their ABO compatibilities. Which is which?
16. Among the genes having the greatest number of alleles are those involved in self-incompatibility in plants. In some cases, hundreds of alleles exist for a

single gene. What types of constraints might exist to set a limit on the number of alleles a gene can have?

17. In the human ABO blood system, the alleles I^A and I^B are dominant to i . What possible phenotypic ratios do you expect from a mating between a type A individual and a type B individual?
18. In screech owls, crosses between red and silver individuals sometimes yield all red; sometimes 1/2 red:1/2 silver; and sometimes 1/2 red:1/4 white:1/4 silver offspring. Crosses between two red owls yield either all red, 3/4 red:1/4 silver, or 3/4 red:1/4 white offspring. What is the mode of inheritance?
19. A premed student, Steve, plans to marry the daughter of the dean of nursing. The dean's husband was sterile, and the daughter was conceived by artificial insemination. Steve's father puts pressure on Steve to marry someone else. Having served as an anonymous sperm donor, he is concerned that Steve and his fiancé may be half brother and sister. Given the following information, deduce whether Steve and his fiancé could be related. (The MN and Ss systems are two independent, codominant blood-type systems.)

	Blood Type
Dean	A, MN, Ss
Her daughter	O, M, S
Steve's father	A, MN, Ss
Steve	O, N, s
Steve's mother	B, N, s

INDEPENDENT ASSORTMENT

20. Mendel self-fertilized dihybrid plants ($RrYy$) with round and yellow seeds and got a 9:3:3:1 ratio in the F_2 generation. As a test of Mendel's hypothesis of independent assortment, predict the kinds and numbers of progeny produced in testcrosses of these F_2 offspring.
21. Four o'clock plants have a gene for color and a gene for height with the following phenotypes:
- | | |
|---------------------|----------------------------|
| RR : red flower | TT : tall plant |
| Rr : pink flower | Tt : medium height plant |
| rr : white flower | tt : dwarf plant |
- Give the proportions of genotypes and phenotypes produced if a dihybrid plant is self-fertilized.
22. A particular variety of corn has a gene for kernel color and a gene for height with the following phenotypes:
- | | |
|------------------------------|---------------------------|
| CC , Cc : purple kernels | TT : tall stem |
| cc : white kernels | Tt : medium height stem |
| | tt : dwarf stem |
- Give the proportions of genotypes and phenotypes produced if a dihybrid plant is selfed.

23. To determine the genotypes of the offspring of a cross in which a corn trihybrid ($Aa Bb Cc$) was selfed, a geneticist has three choices. He or she can take a sample of the progeny and (a) self-fertilize the individual plants, (b) testcross the plants, or (c) cross the individuals with a trihybrid (backcross). Which method is preferable?
24. In figure 2.17, the F_2 phenotypic ratio is 3:1:3:1:6:2. What are the phenotypic segregation ratios for each blood system (AB, Rh) separately? Are they segregating properly? What phenotypic ratio in the F_2 generation would indicate interference with independent assortment?
25. Assume that Mendel looked simultaneously at four traits of his pea plants (and each trait exhibited dominance). If he crossed a homozygous dominant plant with a homozygous recessive plant, all the F_1 offspring would be of the dominant phenotype. If he then selfed the F_1 plants, how many different types of gametes would these F_1 plants produce? How many different phenotypes would appear in the F_2 generation? How many different genotypes would appear? What proportion of the F_2 offspring would be of the fourfold recessive phenotype?
26. A geneticist crossed two corn plants, creating an F_1 decahybrid (ten segregating loci). He then self-fertilized this decahybrid. How many different kinds of gametes did the F_1 plant produce? What proportion of the F_2 offspring were recessive homozygotes? How many different kinds of genotypes and phenotypes were generated in the F_2 offspring? What would your answer be if the geneticist testcrossed the decahybrid instead?
27. Consider the following crosses in pea plants and determine the genotypes of the parents in each cross. Yellow and green refer to seed color; tall and short refer to plant height.

Cross	Progeny			
	Yellow, Tall	Yellow, Short	Green, Tall	Green, Short
a. Yellow, tall $\times$ yellow, tall	89	31	33	10
b. Yellow, short $\times$ yellow, short	0	42	0	15
c. Green, tall $\times$ yellow, short	21	20	24	22

28. A brown-eyed, long-winged fly is mated with a red-eyed, long-winged fly. The progeny are

51 long, red	18 short, red
53 long, brown	16 short, brown

What are the genotypes of the parents?

29. True-breeding flies with long wings and dark bodies are mated with true-breeding flies with short wings and tan bodies. All the F_1 progeny have long wings and tan bodies. The F_1 progeny are allowed to mate and produce:

44 tan, long	14 tan, short
16 dark, long	6 dark, short

What is the mode of inheritance?

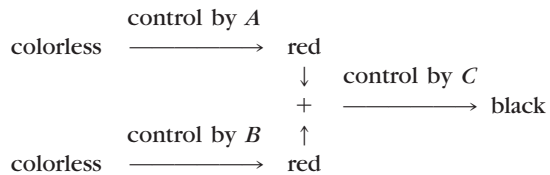
30. In peas, tall (T) is dominant to short (t), yellow (Y) is dominant to green (y), and round (R) is dominant to wrinkled (r). From a cross of two triple heterozygotes, what is the chance of getting a plant that is
- tall, yellow, round?
 - short, green, wrinkled?
 - short, green, round?
31. In corn, the genotype $A- C- R-$ is colored. Individuals homozygous for at least one recessive allele are colorless. Consider the following crosses involving colored plants, all with the same genotype. Based on the results, deduce the genotypes of the colored plants.
- colored $\times aa cc RR \rightarrow 1/2$ colored; $1/2$ colorless
- colored $\times aa CC rr \rightarrow 1/4$ colored; $3/4$ colorless
- colored $\times AA cc rr \rightarrow 1/2$ colored; $1/2$ colorless
32. Consider the following crosses in *Drosophila*. Based on the results, deduce which alleles are dominant and the genotypes of the parents. Orange and red are eye colors; crossveins occur on the wings.

Parents	Progeny			
	Orange, crossveins	Orange, crossveinless	Red, crossveins	Red, crossveinless
a. Orange, crossveins $\times$ orange, crossveins	83	26	0	0
b. Red, crossveins $\times$ red, crossveinless	20	18	65	63
c. Red, crossveinless $\times$ red, crossveins	0	0	74	81
d. Red, crossveins $\times$ red, crossveins	28	11	93	34

33. In *Drosophila melanogaster*, a recessive autosomal gene, *ebony*, produces a dark body color when homozygous, and an independently assorting autosomal

gene, *black*, has a similar effect. If homozygous *ebony* flies are crossed with homozygous *black* flies,

- what will be the phenotype of the F_1 flies?
 - what phenotypes and what proportions would occur in the F_2 generation?
 - what phenotypic ratios would you expect to find in the progeny of the backcrosses of $F_1 \times \text{ebony}$? $F_1 \times \text{black}$?
34. *A*, *B*, and *C* are independently assorting Mendelian factors (genes) controlling the production of black pigment in a rodent species. Alleles of these genes are indicated as *a*, *b*, and *c*, respectively. Assume that *A*, *B*, and *C* act in this pathway:



A black *AA BB CC* individual is crossed with a colorless *aa bb cc* to give black F_1 individuals. The F_1 individuals are selfed to give F_2 progeny.

- What proportion of the F_2 generation is colorless?
 - What proportion of the F_2 generation is red?
35. In a particular *Drosophila* species, there are four strains differing in eye color: wild-type, orange-1, orange-2, and pink. The following matings of true-breeding individuals were performed.

Cross	F_1
wild-type $\times$ orange-1	all wild-type
wild-type $\times$ orange-2	all wild-type
orange-1 $\times$ orange-2	all wild-type
orange-2 $\times$ pink	all orange-2
F_1 (orange-1 $\times$ orange-2) $\times$ pink	1/4 orange-2: 1/4 pink: 1/4 orange-1: 1/4 wild-type

What F_2 ratio would you expect if the F_1 progeny from orange-1 $\times$ orange-2 were selfed?

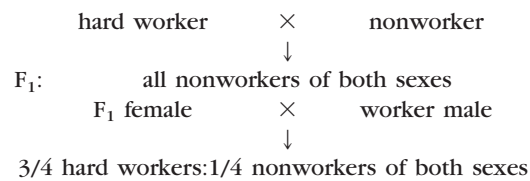
GENOTYPIC INTERACTIONS

36. In a variety of onions, three bulb colors segregate: red, yellow, and white. A plant with a red bulb is crossed to a plant with a white bulb, and all the offspring have red bulbs. When these are selfed, the following plants are obtained:

Red-bulbed	119
Yellow-bulbed	32
White-bulbed	9

What is the mode of inheritance of bulb color, and how do you account for the ratio?

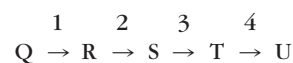
37. When studying an inherited phenomenon, a geneticist discovers a phenotypic ratio of 9:6:1 among offspring of a given mating. Give a simple, genetic explanation for this result. How would you test this hypothesis?
38. You notice a rooster with a pea comb and a hen with a rose comb in your chicken coop. Outline how you would determine the nature of the genetic control of comb type. How would you proceed if both your rooster and hen had rose combs?
39. Suggest possible mechanisms for the epistatic ratios given in table 2.4. Can you add any further ratios?
40. What are the differences among dominance, epistasis, and pleiotropy? How can you determine that pleiotropic effects, such as those seen in sickle-cell anemia, are not due to different genes?
41. You are working with the exotic organism *Phobia laboris* and are interested in obtaining mutants that work hard. Normal phobes are lazy. Perseverance finally pays off, and you successfully isolate a true-breeding line of hard workers. You begin a detailed genetic analysis of this trait. To date you have obtained the following results:



From these results, predict the expected phenotypic ratio from crossing two F_1 nonworkers.

BIOCHEMICAL GENETICS

42. The following is a pathway from substance Q to substance U, with each step numbered:



Which product should build up in the cell and which products should never appear if the pathway is blocked at point 1? At 2? At 3? At 4?

43. The following chart shows the growth (+) or lack of growth (−) of four mutant strains of *Neurospora* with various additives. The additives are in the pathway of niacin biosynthesis. Diagram the pathway and show which steps the various mutants block. Which compound would each mutant accumulate? When you complete this problem, compare your results with figure 2.27. What effect on growth would you observe following a mutation in the pathway of serine biosynthesis?

Additives	Mutants			
	1	2	3	4
Nothing	—	—	—	—
Niacin	+	+	+	+
Tryptophan	+	+	—	—
Kynurenine	+	+	—	—
3-Hydroxyanthranilic acid	+	+	+	—
Indole	—	+	—	—

44. The following shows the growth (+) or lack of growth (—) of various mutants in another biosynthesis pathway. Determine this pathway, the point of blockage for each mutant, and the substrate each mutant accumulates.

Additives	Mutants				
	1	2	3	4	5
Nothing	—	—	—	—	—
A	—	+	+	+	+
B	—	+	—	+	—
C	—	+	—	+	+
D	—	—	—	+	—
E	+	+	+	+	+

45. Maple sugar urine disease is a rare inborn error of human metabolism in which the urine of affected individuals smells like maple sugar.

- If two unaffected individuals have an affected child, what is the probable mode of inheritance of the disease?
- What is the chance that the second child will be unaffected?

CRITICAL THINKING QUESTIONS

- In the shepherd's purse plant, the seed capsule comes in two forms, triangular and rounded. If two dihybrids are crossed, the resulting ratio of capsules is 15:1 in favor of triangular seed capsules. What type of biochemical pathway might generate that ratio?

- Assume Mendel made the cross of two true-breeding plants that differed in all seven traits under study, one with all dominant traits, the other with all recessive traits. What would the ratio of phenotypes be in the F_2 generation?

Suggested Readings for chapter 2 are on page B-1.

3

MITOSIS AND MEIOSIS

STUDY OBJECTIVES

1. To observe the morphology of chromosomes 48
2. To understand the processes of mitosis and meiosis 50
3. To analyze the relationships between meiosis and Mendel's rules 61

STUDY OUTLINE

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Mitosis 52

The Mitotic Spindle 52

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Meiosis 55

Prophase I 56

Metaphase I and Anaphase I 59

Telophase I and Prophase II 59

Meiosis II 60

The Significance of Meiosis 61

Meiosis in Animals 63

Life Cycles 64

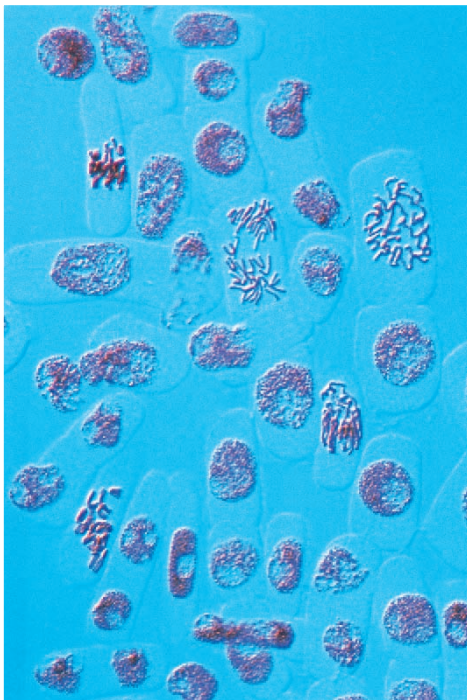
Chromosomal Theory of Heredity 66

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Onion (*Allium cepa*) cells in various stages of mitosis.

(© Andrew Syred/Tony Stone Images.)

The zygote, or fertilized egg of higher organisms, is the starting point of most life cycles. This zygote then divides many times to produce an adult organism. In animals, the adults then produce gametes that combine to start the cycle again. In higher plants, the adult is a sporophyte that produces spores by genetic reduction. These spores develop into gametophytes, which may or may not be independent, and gametophytes produce gametes that fuse to form the zygote (fig. 3.1). (Numerous variations on these themes exist, some of which we will discuss later in this chapter or others.) The process of cell division includes a nuclear and a cytoplasmic component. Nuclear division (**karyokinesis**) has two forms, a nonreductional **mitosis** in which the mother and daughter cells have exactly the same genetic complement, and a reductional **meiosis** in which the products, gametes in animals and spores in higher plants, have approximately half

the genetic material as the parent cell. Halving the amount ensures that, when the gametes recombine, the amount of genetic material in a zygote is the same from generation to generation. The division of the cytoplasm, resulting in two cells from one original cell, is termed **cytokinesis**. In this chapter, we examine the processes of mitosis and meiosis, which allow chromosomes, the gene vehicles, to properly apportion among daughter cells. We will discuss the engineering difficulties these processes pose and the relationship of meiosis to Mendel's rules.

Mendel's work was rediscovered at the turn of the century after being ignored for thirty-four years. One of the major reasons scientists could appreciate it in 1900 was that many of the processes that chromosomes undergo had been described. With those discoveries, a physical basis for genes had been found. That is, chromosomal behavior during gamete formation precisely fits Mendel's predictions for gene behavior during gamete formation. In this chapter, we look at the morphology of chromosomes and their behavior during somatic-cell division and gamete and spore formation.

Modern biologists classify organisms into two major categories: **eukaryotes**, organisms that have true, membrane-bound nuclei, and **prokaryotes**, organisms that lack true nuclei (table 3.1). Bacteria and blue-green algae are prokaryotes. All other organisms are eukaryotes. In most prokaryotes, the genetic material is a circle of double-stranded DNA (deoxyribonucleic acid) with some associated proteins; ancillary circles of double-stranded DNA called plasmids are also found frequently (see chapters 13 and 17). In eukaryotes, the genetic material, located in the nucleus (fig. 3.2), is linear, double-stranded DNA highly complexed with protein (**nucleoprotein**). In this chapter, we concentrate on the nuclear division processes of eukaryotes.

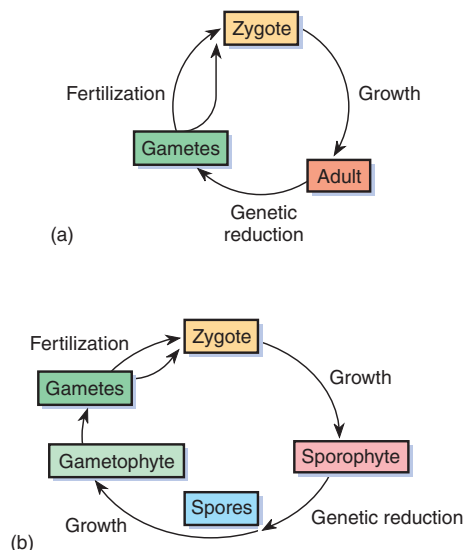


Figure 3.1 Generalized life cycle of (a) animals and (b) plants.

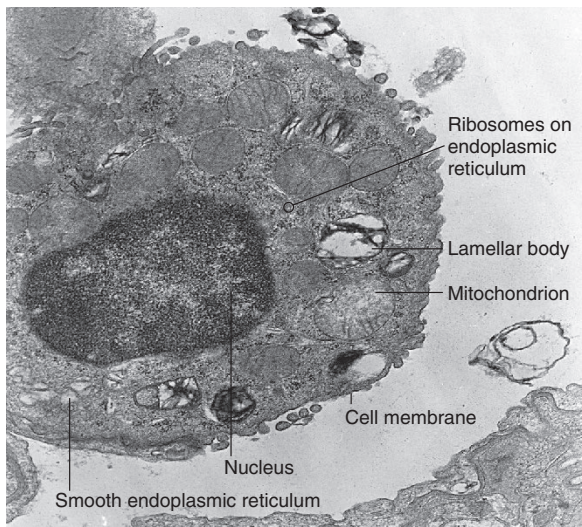
Table 3.1 Differences Between Prokaryotic and Eukaryotic Cells

	Prokaryotic Cells	Eukaryotic Cells
Taxonomic groups	Bacteria	All plants, fungi, animals, protists
Size*	Usually less than 5 μm in greatest dimension	Usually greater than 5 μm in smallest dimension
Nucleus	No true nucleus, no nuclear membrane	Nuclear membrane
Genetic material	One circular molecule of DNA, little protein	Linear DNA molecules complexed with histones
Mitosis and meiosis	Absent	Present

* See table 3.2 on page 48.

Table 3.2 Metric Units of Linear Measurement

Unit	Abbreviation	Size
meter	m	39.37 U.S. inches
centimeter	cm	10^{-2} meter
millimeter	mm	10^{-3} meter
micrometer	μm	10^{-6} meter
nanometer	nm	10^{-9} meter
Angstrom	$\text{\AA}$	10^{-10} meter

**Figure 3.2** Mouse lung cell magnified 4,270x. (Courtesy of Wayne Rosenkrans.)

CHROMOSOMES



Chromosomes were discovered by C. von Nägeli in 1842. The term **chromosome**, which W. Waldeyer coined in 1888, means “colored body.” Von Nägeli discovered chromosomes after staining techniques were developed that made them visible. The nucleoprotein material of the chromosomes is referred to as **chromatin**. When diffuse, chromatin is referred to as **euchromatin**; when condensed and readily visible, as **heterochromatin**.

Although all eukaryotes have chromosomes, in the **interphase** between divisions, they are spread out or diffused throughout the nucleus and are usually not identifiable. Each chromosome, with very few exceptions, has a distinct attachment point for fibers (**microtubules**) that make up the mitotic and meiotic spindle apparatuses. The attachment point occurs at a constriction in

the chromosome termed the **centromere**, which is composed of several specific DNA sequences (see chapter 15). The **kinetochore** is the proteinaceous structure on the surface of the centromere to which microtubules of the spindle attach. Chromosomes can be classified according to whether the centromere is in the middle of the chromosome (**metacentric**), at the end of the chromosome (**telocentric**), very near the end of the chromosome (**acrocentric**), or somewhere in between (**subtelocentric** or **submetacentric**; figs. 3.3 and 3.4). For any particular chromosome, the position of the centromere is fixed. In various types of preparations, dark bands (**chromomeres**) are visible (see chapter 15).

Most higher eukaryotic cells are **diploid**; that is, all their chromosomes occur in pairs. One member of each pair came from each parent. **Haploid** cells, which include the reproductive cells (gametes), have only one copy of each chromosome. In the diploid state, members of the same chromosome pair are referred to as **homologous chromosomes** (homologues); the two make up a homologous pair.

The total chromosomal complement of a cell, the **karyotype**, can be photographed during mitosis and rearranged in pairs to make a picture called a karyotype or **idiogram** (fig. 3.5). From the idiogram it is possible to see whether the chromosomes have any abnormalities and to identify the sex of the organism. As you can see from figure 3.5, all of the homologous pairs are made up of identical partners, and are thus referred to as **homomorphic chromosome pairs**. A potential exception is the sex chromosomes, which in some species are of un-

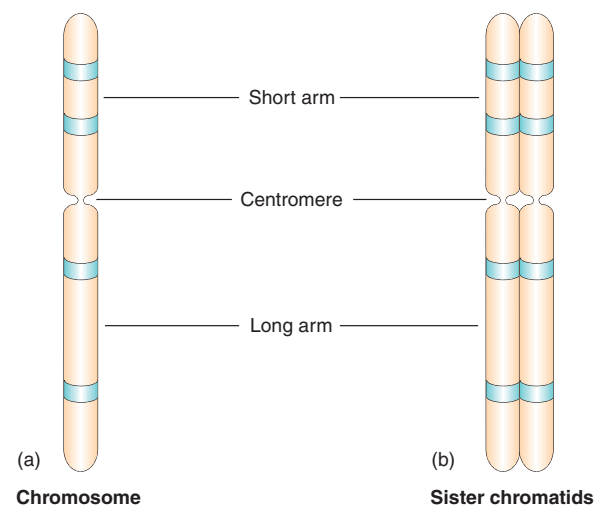
**Figure 3.3** (a) Submetacentric chromosome and (b) submetacentric chromosome in mitosis. The chromosome is best seen after it has duplicated but before the identical halves (sister chromatids) separate.



Figure 3.4 (a) Metacentric, (b) submetacentric, and (c) acrocentric chromosomes in human beings. Except in telocentric chromosomes, the centromere divides the chromosome into two arms. (Reproduced courtesy of Dr. Thomas G. Brewster, Foundation for Blood Research, Scarborough, Maine.)

equal size and are therefore called a **heteromorphic chromosome pair**.

The number of chromosomes individuals of a particular species possess is constant. Some species exist mostly in the haploid state or have long haploid intervals in their life cycle. For example, pink bread mold, *Neurospora crassa*, a fungus, has a chromosome number of seven ($n = 7$) in the haploid state. Its diploid number is, of course, fourteen ($2n = 14$). The diploid chromosome numbers of several species appear in table 3.3.

In eukaryotes, two processes partition the genetic material into offspring, or daughter, cells. One is the simple division of one cell into two. In this process, the two daughter cells must each receive an exact copy of the genetic material in the parent cell. The cellular process is simple cell division, and the nuclear process accompanying it is mitosis. In the other partitioning process, the genetic material must precisely halve so that fertilization will restore the diploid complement. The cellular process is gamete formation in animals and spore formation in higher plants, and the nuclear process is meiosis. The term *mitosis* comes from the Greek word for “thread,” referring to a chromosome. The term *meiosis* comes from the Greek meaning “to lessen.”

Chromosomes separate in both processes of nuclear division. The division of the cytoplasm of the cell, *cytokinesis*,

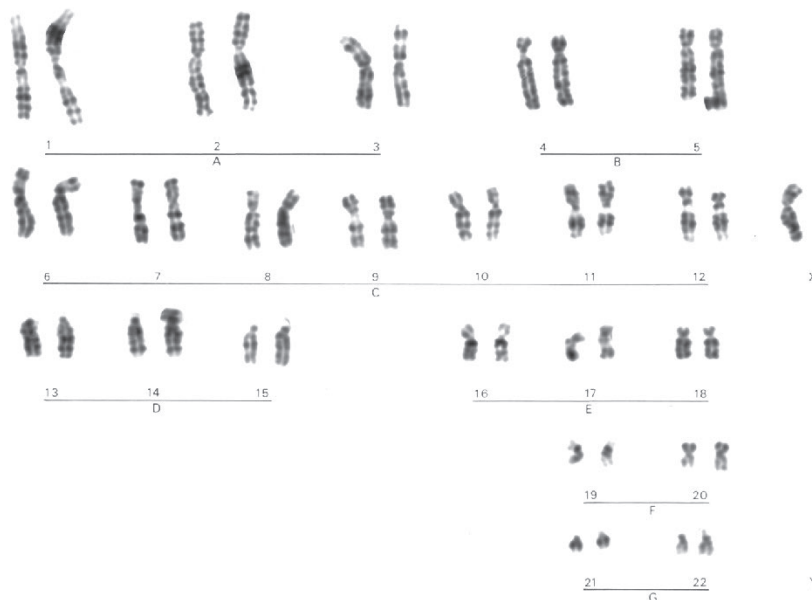


Figure 3.5 Idiogram or karyotype of a human female (two X chromosomes, no Y chromosome). A male would have one X and one Y chromosome. The chromosomes are grouped into categories (A–G, X, Y) by length and centromere position. Similar chromosomes are often distinguished by their chromomeres. (Reproduced courtesy of Dr. Thomas G. Brewster, Foundation for Blood Research, Scarborough, Maine.)

Table 3.3 Chromosome Number for Selected
Species

Species	$2n$
Human being (<i>Homo sapiens</i>)	46
Garden pea (<i>Pisum sativum</i>)	14
Fruit fly (<i>Drosophila melanogaster</i>)	8
House mouse (<i>Mus musculus</i>)	40
Roundworm (<i>Ascaris</i> sp.)	2
Pigeon (<i>Columba livia</i>)	80
Boa constrictor (<i>Constrictor constrictor</i>)	36
Cricket (<i>Gryllus domesticus</i>)	22
Lily (<i>Lilium longiflorum</i>)	24
Indian fern (<i>Ophioglossum reticulatum</i>)	1,260

Note: $2n$ is the diploid complement. The fern has the highest known diploid chromosome number.

is less organized. In animals, a constriction of the cell membrane distributes the cytoplasm. In plants, the growth of a cell plate accomplishes the same purpose.

THE CELL CYCLE

The continuity of life depends on cells growing, replicating their genetic material, and then dividing, a process called the **cell cycle** (fig. 3.6). Although cells usually divide when they have doubled in volume, the control of this process is very complex and precise. Not only do all the steps have to occur in sequence, but the cell must also “know” when to proceed and when to wait. Continuing at inappropriate moments—for example, before the DNA has replicated or when the chromosomes or spindle are damaged—could have catastrophic consequences to a cell or a whole organism. Numerous stops occur during the cycle to assess whether the next step should proceed.

Early research into the cell cycle involved fusing cells in different stages of the cycle (such as the G_1 , S, and G_2 phases; see fig. 3.6) to determine whether the cytoplasmic components of one cell would affect the behavior of the other. Results of these experiments led to the discovery of a protein complex called the **maturation-promoting factor (MPF)** because of its role in causing oocytes to mature. It is now also referred to as the **mitosis-promoting factor** since it initiates the mitosis phase of the cell cycle. Further research has shown that MPF is made of two proteins, one that oscillates in quantity during the cell cycle and one whose quantity is con-

stant. The oscillating component is referred to as **cyclin**; the constant gene product is an enzyme controlled by the *cdc2* gene (*cdc* stands for cell division cycle) called Cdc2p. Cdc2p is a kinase, an enzyme that phosphorylates other proteins, transferring a phosphate group from ATP to an amino acid of the protein it is acting on. (Phosphorylation controls many of the processes in mitosis and in metabolism in general; for example, the nuclear membrane begins to break down when its subunits are phosphorylated.) Because the Cdc2p kinase works when combined with cyclin, it is referred to as a **cyclin-dependent kinase (CDK)**. Several of these kinase-cyclin combinations control stages of the cell cycle; the cyclin of the mitosis-promoting factor is called cyclin B. In general, cyclin-dependent kinases are regulated by phosphorylation and dephosphorylation, cyclin levels, and activation or deactivation of inhibitors.

Normally, Cdc2p remains at high levels in the cell but does not initiate mitosis for two reasons. First, phosphate groups block its active site, the place on the enzyme that actually does the phosphorylating. Second, the enzyme can only function when it combines with a molecule of cyclin B, the protein that oscillates during the cell cycle. Cyclin B is at very low levels when mitosis ends. During ensuing cell growth, numbers of cyclin B molecules increase, combining with Cdc2p proteins until a critical quantity is reached. However, Cdc2p-cyclin B complexes are still not active. That requires the product of another gene to dephosphorylate the Cdc2p-cyclin B complex. At that point, the Cdc2p-cyclin B complex goes into action, initiating the changes that begin mitosis (fig. 3.7). Presumably the cell is ready for mitosis at that point, having

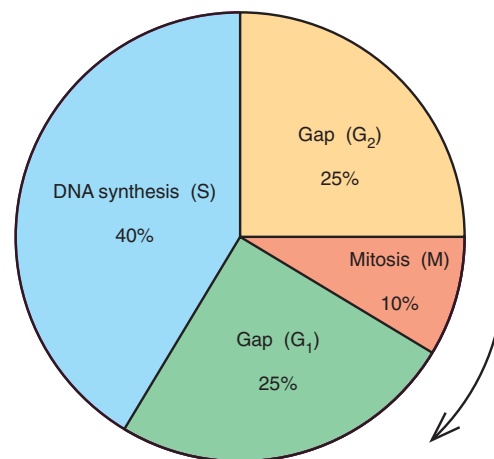


Figure 3.6 Cell cycle in the broad bean, *Vicia faba*. Total time in the cycle is under twenty hours. The DNA content of the cell doubles during the S phase and is then reduced back to its original value by mitosis.

gone through G₁, S, and G₂ phases (which we will discuss in detail later in the chapter).

Once mitosis has been initiated, cyclin B, along with other proteins that have served their purpose by this point in the cell cycle, breaks down with the help of a protein complex called the **anaphase-promoting complex (APC)**, also called the **cyclosome**. The cyclosome works by attaching a **ubiquitin** molecule to the proteins that are to be broken down. (Ubiquitin is a polypeptide of 76 amino acids; it directs the attached protein into a breakdown pathway discussed in chapter 16.) Cdc2p is then phosphorylated to block its active site. The cell now completes mitosis and enters G₁; quantities of cyclin B are very low, and virtually no functioning Cdc2p-cyclin B remains (fig. 3.7). Thus, active Cdc2p is the kinase that controls the initiation of mitosis.

Some points in the cell cycle, such as the initiation of mitosis, can be delayed until all necessary conditions are

in place. These **checkpoints** allow the cell to make sure that various events have been “checked off” as completed before the next phase begins. **Surveillance mechanisms** that involve dozens of proteins, many just discovered, oversee these checkpoints. In the cell cycle, three checkpoints involve cyclin-dependent kinases; each has its own specific cyclin that initiates either the G₁, S, or mitosis phase. In addition, other checkpoints that don’t involve cyclin-dependent kinases occur at other transition phases in the cell cycle.

Cell cycle control is of particular interest because the cell cycle routinely halts if there is genetic damage, giving the cell a chance to repair the damage before committing to cell division. If the damage is too extreme, the cell can enter a programmed cell death sequence, discussed in chapter 16. If these mechanisms fail, cancer may result. The genetic control of the cell cycle is one of the most active areas of current research.

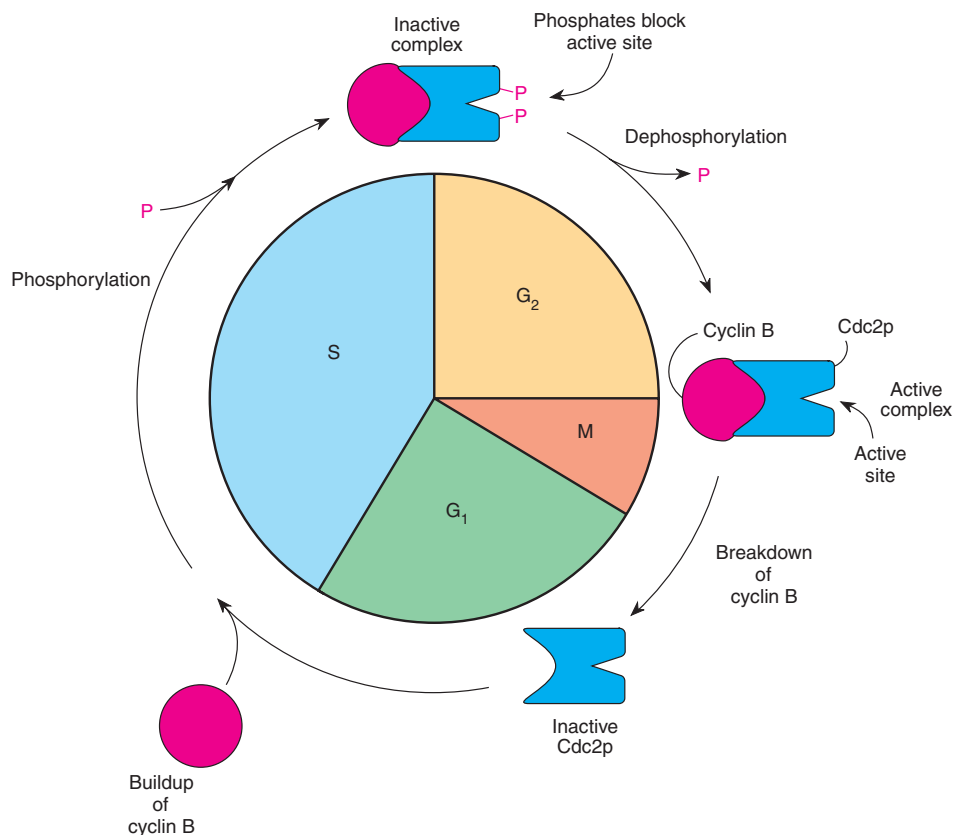


Figure 3.7 The proteins Cdc2p (CDK1) and cyclin B combine to form the maturation-promoting (or mitosis-promoting) factor. During mitosis, cyclin B is broken down. During G₁ and S phases, cyclin B builds up and combines with Cdc2p, which is then phosphorylated at the active site to render it inactive. Dephosphorylation, a process that begins to take place only after DNA replication is finished, produces an active maturation-promoting factor.

MITOSIS

Consider the engineering problem that mitosis must solve. Identical **chromatids**, called **sister chromatids**, the result of chromosomal replication, must separate so that each goes into a different daughter cell (see fig. 3.3). These chromatids are the visible manifestation of the chromosomal replication that has taken place in the S phase of the cell cycle. The chromatids are initially held together; each will be called a chromosome when it separates and becomes independent. Each of the two daughter cells then ends up with a chromosome complement identical to that of the parent cell. Mitosis is nature's elegant process to achieve that end—surely an engineering marvel.

Mitosis is a continuous process. However, for descriptive purposes, we can break it into four stages: **prophase**, **metaphase**, **anaphase**, and **telophase** (Greek: *pro-*, before; *meta-*, mid; *ana-*, back; *telo-*, end). Replication (duplication) of the genetic material occurs during the S phase of the cell cycle (see fig. 3.6). The timing of the four stages varies from species to species, from organ to organ within a species, and even from cell to cell within a given cell type.

The Mitotic Spindle

The process of mitosis involves an apparatus called the **spindle**. This structure is composed of microtubules, hollow cylinders made of protein subunits; each subunit is composed of one molecule of α tubulin and one of β tubulin; and each tubulin is the product of a different gene. (The spindle is named for the rounded rods, tapered at each end, once commonly used to hold yarn or thread.) Microtubules provide shape and structure to a eukaryotic cell as well as allow the cell to move its internal components and to move the cell itself with cilia and flagella. Motion occurs as the microtubules slide past each other, a vesicle of some kind slides along the microtubules, and the microtubules shorten. Two proteins make up the microtubule motors that allow motion: **kinesin** and **dynein**. Scientists have studied microtubules through protein chemistry, through mutant organisms, and through innovative methods such as by coupling tubulin subunits with fluorescing dyes to observe the microtubules in action.

Microtubules are in a dynamic equilibrium, with subunits constantly being added or removed at both ends. On any microtubule, more activity occurs at one end than the other. The more active end of the tubule is called the plus end, the less active end the minus end (fig. 3.8). Both ends may be adding or removing subunits, or the plus end may be adding while the minus end is removing subunits. Generally, dynein causes movement toward the minus end, whereas kinesin causes movement toward the plus end of a microtubule, although exceptions exist.

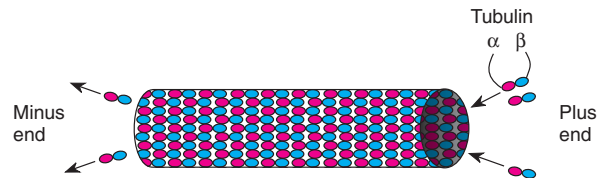


Figure 3.8 Microtubules are hollow tubes made of α and β tubulin subunits that are constantly being added or removed.

Microtubules are formed from active centers called **microtubule organizing centers**. **Centrioles**, composed of two cylinders—themselves composed of microtubules—are microtubule organizing centers for cilia and flagella. Under those circumstances, the centrioles are referred to as **basal bodies**. The centrioles were also originally believed to organize spindles. However, for most organisms, the microtubule organizing center is called the **centrosome**. In some organisms, such as fungi, a different cell organelle, the **spindle pole body**, serves this function. In most animals, the centrosome contains a centriole (fig. 3.9). However, the centriole is absent in most higher plants. Moreover, innovative ex-

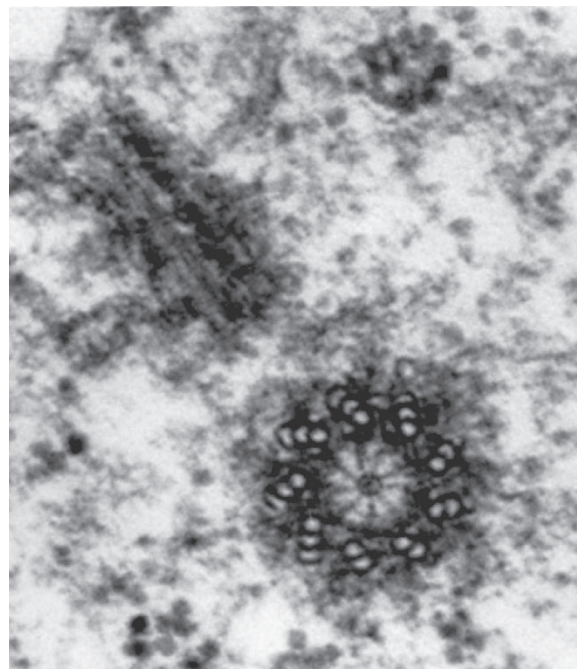


Figure 3.9 A centriole is composed of two barrels at right angles to each other. Each barrel is composed of nine tripartite units and a central cartwheel. Each of the three parts of a tripartite unit is a microtubule. Magnification 111,800 $\times$. (Reproduced from *The Journal of Cell Biology*, 1968, Vol. 37, p. 381, by copyright permission of The Rockefeller University Press.)

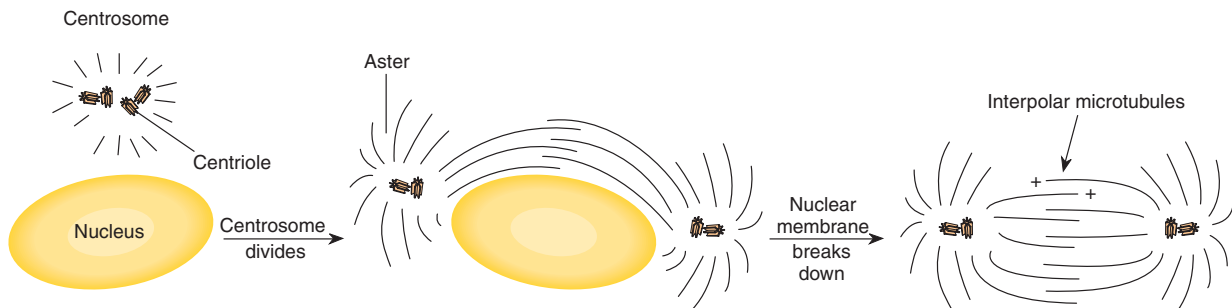


Figure 3.10 Early in mitosis, the centrosome divides, and the separating halves move to opposite poles of the cell. This creates a spindle in the middle of the cell after the nuclear membrane breaks down.

periments that removed the centrioles from cells that normally had them demonstrated that the centriole is not necessary for spindle formation. So, although we used to believe that the centriole formed the spindle in many organisms, we now know that the spindle is usually organized around the centrosome, which can function in this capacity without a centriole.

The centriole, when present, replicates during the S and G₂ phases. When mitosis begins, the centrosome divides and moves to opposite poles of the cell, around the nucleus (fig. 3.10). The centrosomes trail microtubules, forming the spindle, that at this point begin at each centrosome and overlap in the middle of the cell. These are called **interpolar microtubules**. Microtubules also spread out from the centrosome in the opposite direction from the spindle itself, forming an **aster** (see fig. 3.10). The minus ends of microtubules emanate from the centrosome and the plus ends overlap in the middle of the cell. A third form of tubulin, γ tubulin, is needed to begin the formation of a microtubule.

Prophase

This stage of mitosis is characterized by the formation of the spindle and a shortening and thickening of the chromosomes so that individual chromosomes become visible. (We will discuss details of the molecular structure of the eukaryotic chromosome and the processes of coiling and shortening in chapter 15.) At this time also, the nuclear envelope (membrane) disintegrates and the **nucleolus** disappears (fig. 3.11). The nucleolus is a darkly stained body in the nucleus that is involved in ribosome construction and that forms around a **nucleolar organizer** locus on one of the chromosome pairs. The number of nucleoli varies in different species, but in the simplest case there are two nucleolar organizers per nucleus, one each on the two members of a homologous pair of chromosomes. Nucleoli re-form after mitosis.

As prophase progresses, each chromosome is composed of two identical (sister) chromatids (see fig. 3.3); the

chromosomes continue to shorten and thicken. The centromeres have already divided, and no new DNA synthesis is needed for the process to be completed. At this point, the sister chromatids are kept together by a complex, called **cohesin**, made up of at least four different proteins.

Spindle fibers are initially nucleated at the centrosome and grow outward into the cytoplasm (fig. 3.12). Some of these fibers “capture” a kinetochore, the proteinaceous complex at the centromere of each sister chromatid; these fibers are called **kinetochore microtubules**. At first, one kinetochore or the other randomly attaches to a spindle fiber. As the microtubules further move the chromosomes and as new microtubules attach and old microtubules break, each sister kinetochore eventually attaches to microtubules emanating from different poles. This ensures that sister chromatids move to opposite poles during anaphase. The number of microtubules that attach to each kinetochore differs in different species. It seems that 1 attaches to each kinetochore in yeast, 4 to 7 attach to each kinetochore in the cells of a rat fetus, and 70 to 150 attach in the plant *Haemanthus* (see fig. 3.12).

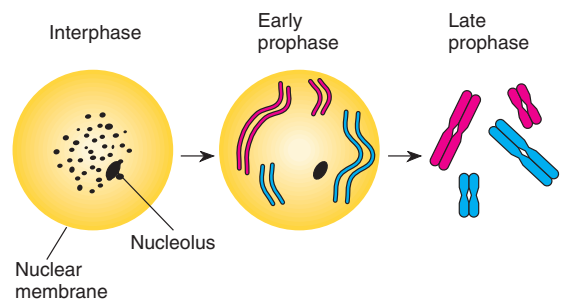


Figure 3.11 Nuclear events during interphase and prophase of mitosis. In this cell, $2n = 4$, consisting of one pair of long and one pair of short metacentric chromosomes. Maternal chromosomes are red; paternal chromosomes are blue. Note that each chromosome consists of two chromatids when the cell enters mitosis.

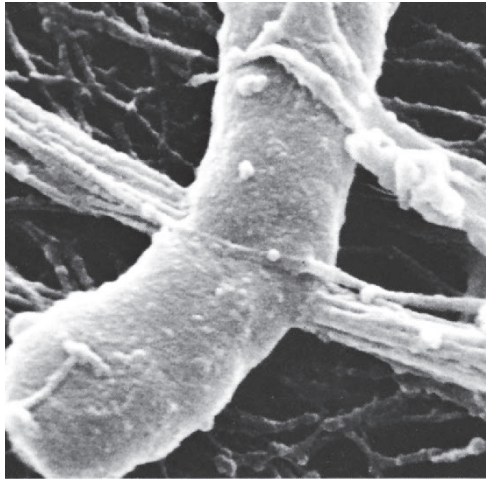


Figure 3.12 Scanning electron micrograph of the centromeric region of a metaphase chromosome from the plant *Haemanthus katherinae*. Spindle fiber bundles on either side of the centromere extend in opposite directions. A fiber not connected to the kinetochore is visible lying over the centromere. These fibers are 60 to 70 nm in diameter. (Waheed K. Heneen, "The centromeric region in the scanning electron microscope," *Hereditas*, 97 (1982): 311–14. Reproduced by permission.)

Metaphase

During metaphase, the chromosomes move to the equator of the cell. With the attachment of spindle fibers and the completion of the spindle itself, the chromosomes jockey into position in the equatorial plane of the spindle, called the **metaphase plate**. This happens as kinetochore microtubules exert opposing tension on the two sister kinetochores. Alignment of the chromosomes on this plate marks the end of metaphase (fig. 3.13).

Anaphase

During anaphase, the sister chromatids separate and move toward opposite poles on the spindle. The physical separation of the sister chromatids and their movement to opposite poles are two separate activities. Chromatid separation represents a checkpoint in the process of mitosis; a surveillance mechanism will not allow the process to continue until all chromosomes are lined up on the metaphase plate with their sister kinetochores held by microtubules from opposite poles. The surveillance mechanism somehow checks the physical tension the spindle fibers exert on a pair of sister chromatids; an unpaired chromatid can delay or stop the process. Initially, an inhibitory protein called **securin** binds an enzyme called **separin** that can break down cohesin, the complex holding the chromatids together. At the correct

moment, the cyclosome ubiquitinates the inhibitor, causing it to break down and freeing the separin to break down cohesin. This liberates the sister chromatids from each other (and is the instant when chromatids become chromosomes).

The spindle then separates the sister chromatids in two stages, called **anaphase A** and **anaphase B**. In anaphase A, the chromosomes move toward the poles (fig. 3.14). During this process, the kinetochore itself acts as a microtubule motor, disassembling microtubules as it moves down them, pulling the chromosomes along (fig. 3.15). Thus, metacentric chromosomes appear V-shaped (as in fig. 3.15), submetacentrics appear J-shaped, and telocentrics appear rod-shaped. In anaphase B, the spindle itself elongates as overlapping inter-polar microtubules slide apart. The general elongation of the spindle pulls the chromosomes apart.

Telophase

At the end of anaphase (fig. 3.16), the separated sister chromatids (now full-fledged chromosomes) have been pulled to opposite poles of the cell. The cell now reverses the steps of prophase to return to the interphase state (fig. 3.17). The chromosomes uncoil and begin to direct protein synthesis. A nuclear envelope re-forms around each set of chromosomes, nucleoli re-form, and cytokinesis takes place. The spindle breaks down into tubulin subunits; a residual of microtubules remains at

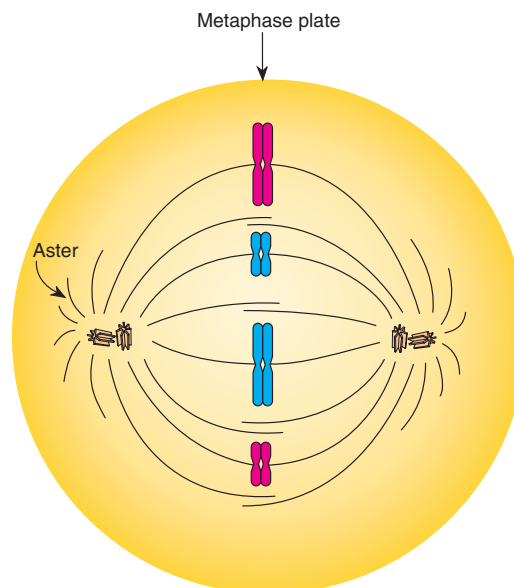


Figure 3.13 Metaphase of mitosis. In this cell, $2n = 4$. Maternal chromosomes are red; paternal chromosomes are blue.